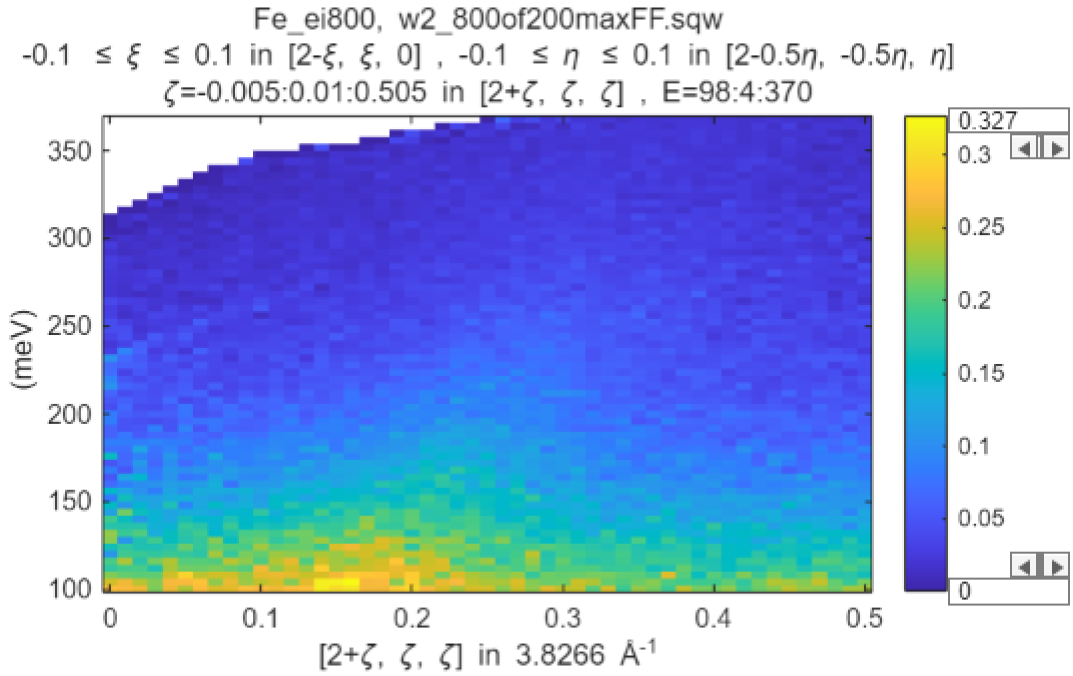
```
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
GHFFfitfile = fullfile(scuts_path,'e800GPffCuts.mat');
if ~exist('GHFFfitData','var')
    ld = load(GHFFfitfile);
    GHFFfitData = ld.fit_src_struc;
end
```

```
mi = maps_instrument(787,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);

cut_names = fieldnames(GHFFfitData);
for i=1:numel(cut_names)
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_instrument(mi);
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_sample(sample);
end
```

```
dir_name  = "GP";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

if ~exist('fit_reEi800GP','var')
    res_name = 'FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat';
    if isfile(res_name)
        ld = load(res_name);
        fit_reEi800GP = ld.fit_reEi800GP;
    else
        fit_reEi800GP = struct();
    end
end


for i=1:numel(cut_names)
    acolor k
    fprintf('******************************\n')
    fprintf('**** CUT: %s\n',cut_names{i});
    fprintf('******************************\n')

    cut_to_work = GHFFfitData.(cut_names{i});
    if isfield(fit_reEi800GP,cut_names{i})
        continue;
    end
    plot(cut_to_work); keep_figure;
    all_en = cut_to_work.data.p{2};
    en_range = min_max(all_en);
    cut_en = 100+half_dE:10:en_range(2)-half_dE;
```

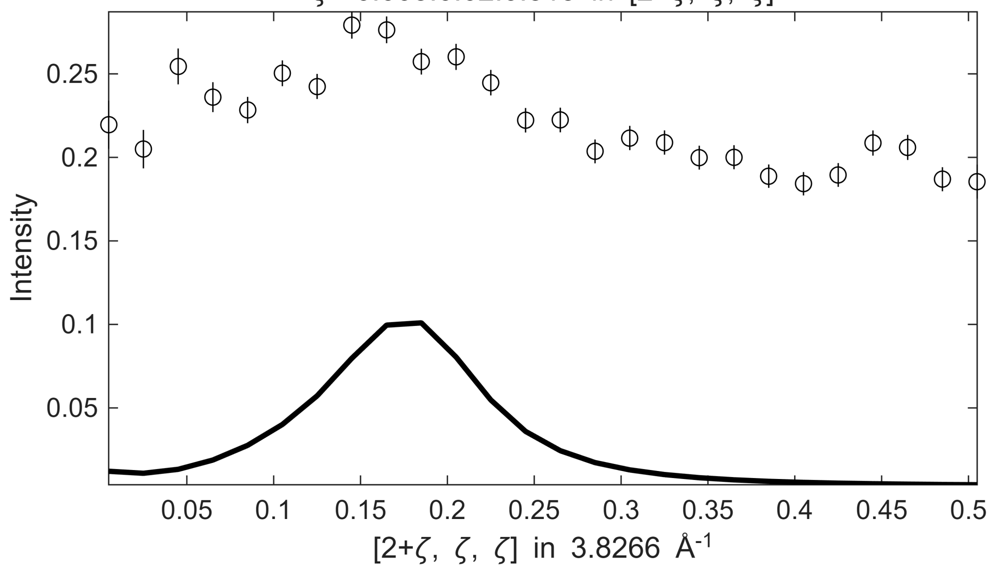
```
    [fr_800const_bg,res_name] = fit_multicuts_along_direction(...
        {cut_to_work},'FitEn_Ei800ff_1D',dir_name, ...
        1,cut_en,dE_step,half_dE,true);
    fit_reEi800GP.(cut_names{i}) = fr_800const_bg;

    acolor g
    bg_data = extract_bg_par(fr_800const_bg.all_fit_par,[1,2]);
    % original slope parameters

pd(bg_data(1));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure; ly 0 0.5;

pd(bg_data(2));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure

    save(res_name,'fit_reEi800GP');
end
```

```
******************************
**** CUT: Ei800_off110minFFGP
******************************
******************************
**** CUT: Ei800_off200maxFFGP
******************************
```



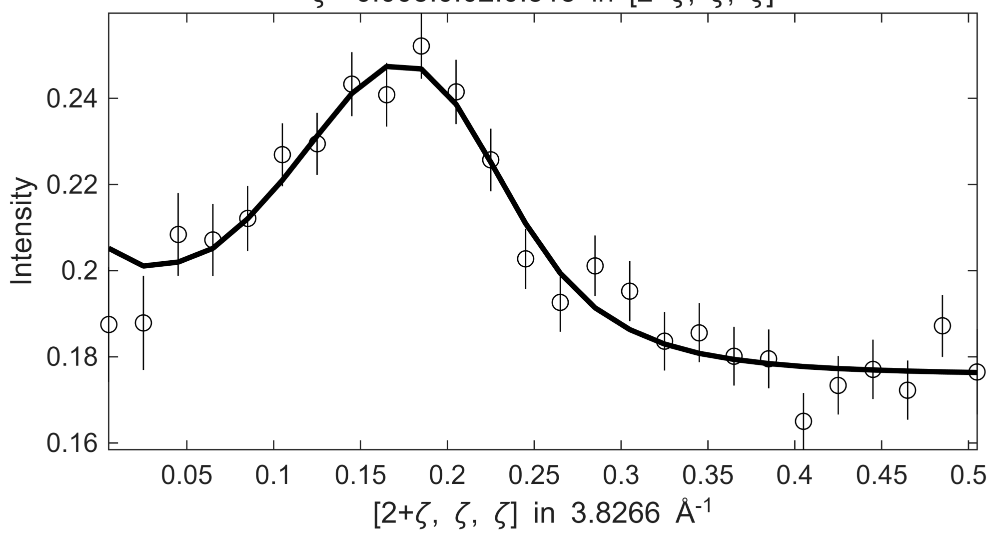

```
******************************
**** En = 105±5 going to 365
******************************
```

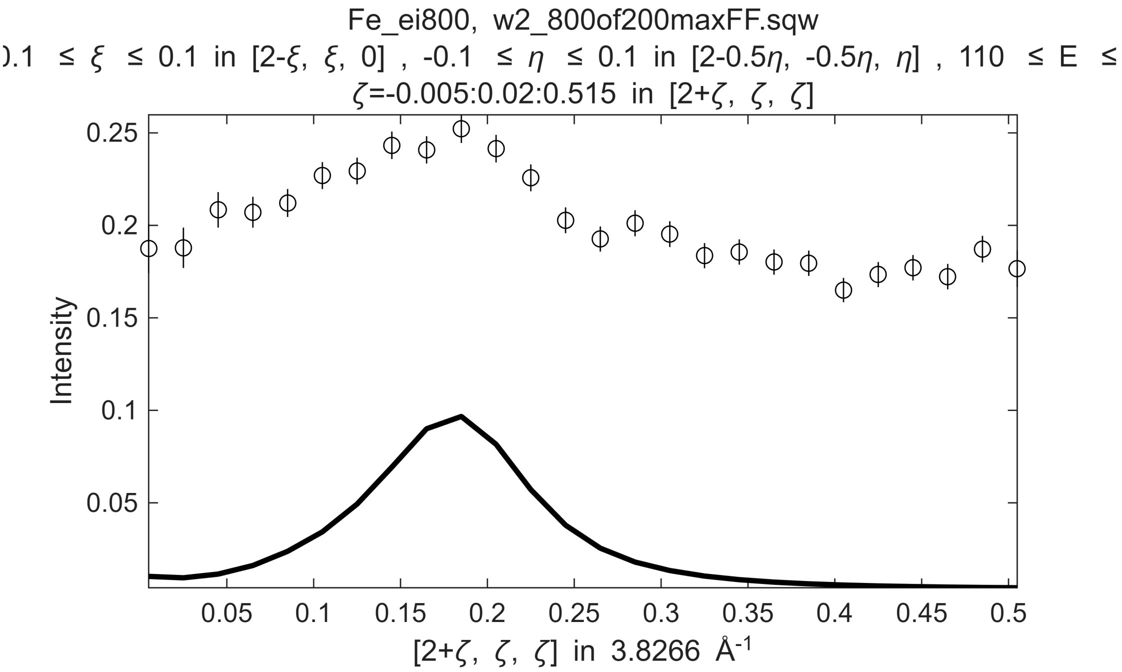
Fe_ei800, w2_800of200maxFF.sqw
).1 ≤ ξ ≤ 0.1 in [2-ξ, ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2-0.5η, -0.5η, η] , 100 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2+ζ, ζ, ζ]



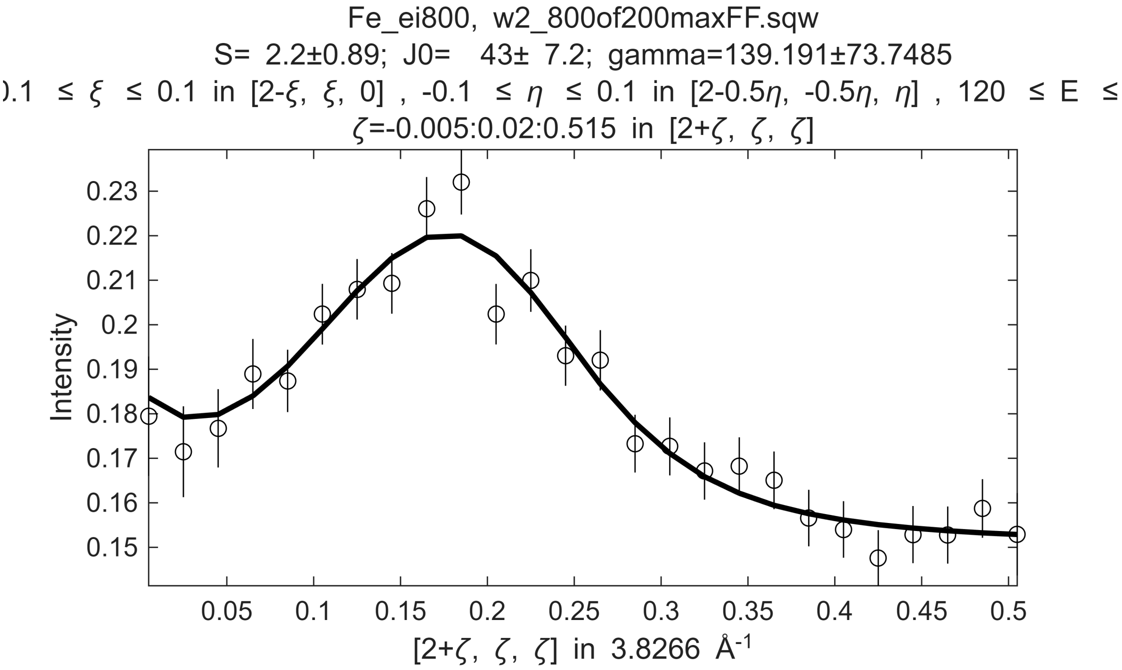

```
*****************************
**** En = 115±5 going to 365
*****************************
```

Fe_ei800, w2_800of200maxFF.sqw
S= 1.1±0.048; J0= 34±0.89; gamma=54.0839±1.70647
).1 ≤ ξ ≤ 0.1 in [2-ξ, ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2-0.5η, -0.5η, η] , 110 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2+ζ, ζ, ζ]

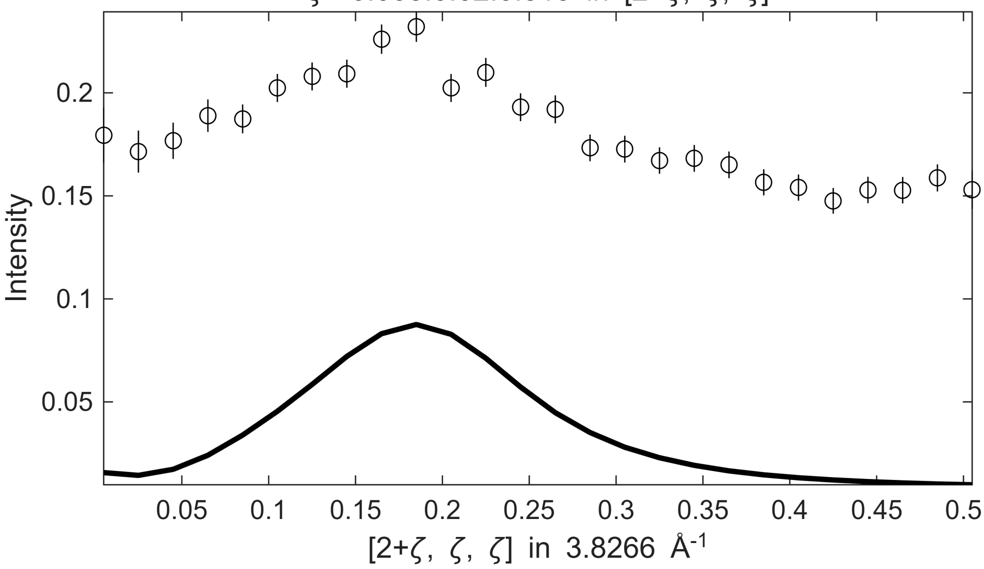





```
*****************************
**** En = 125±5 going to 365
*****************************
```

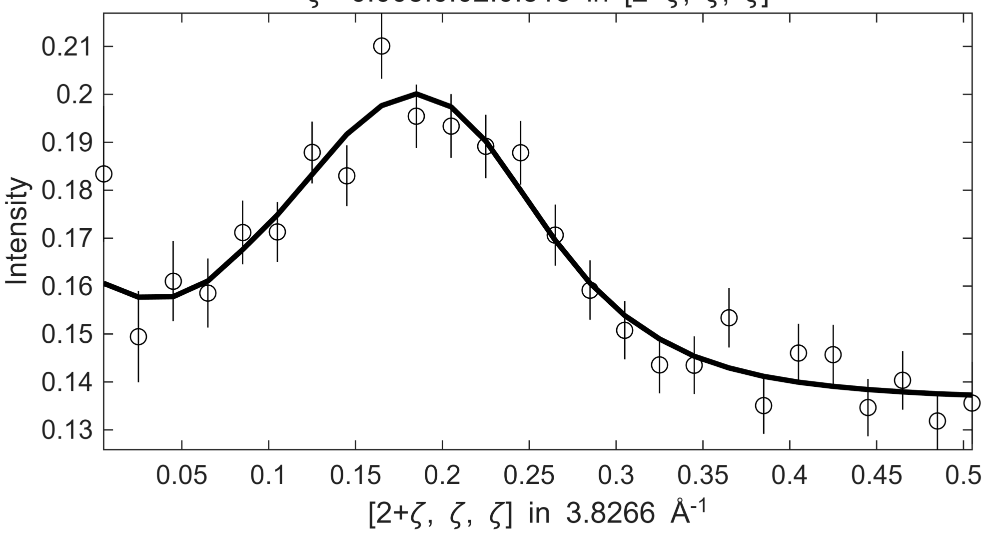


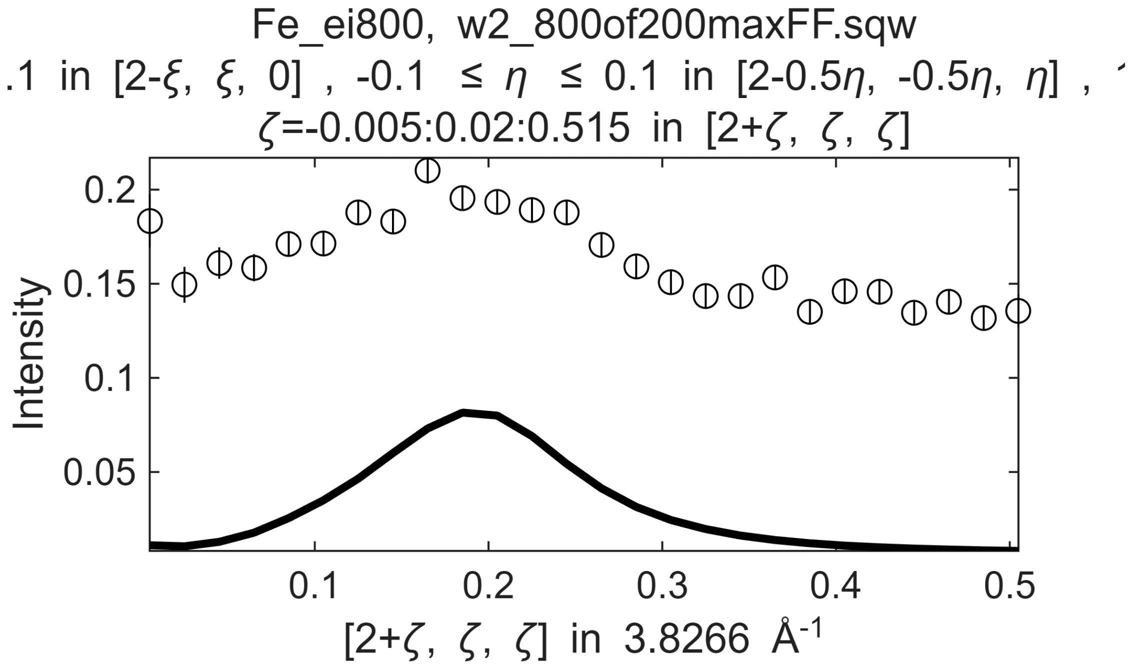
Fe_ei800, w2_800of200maxFF.sqw
).1 ≤ ξ ≤ 0.1 in [2-ξ, ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2-0.5η, -0.5η, η] , 120 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2+ζ, ζ, ζ]



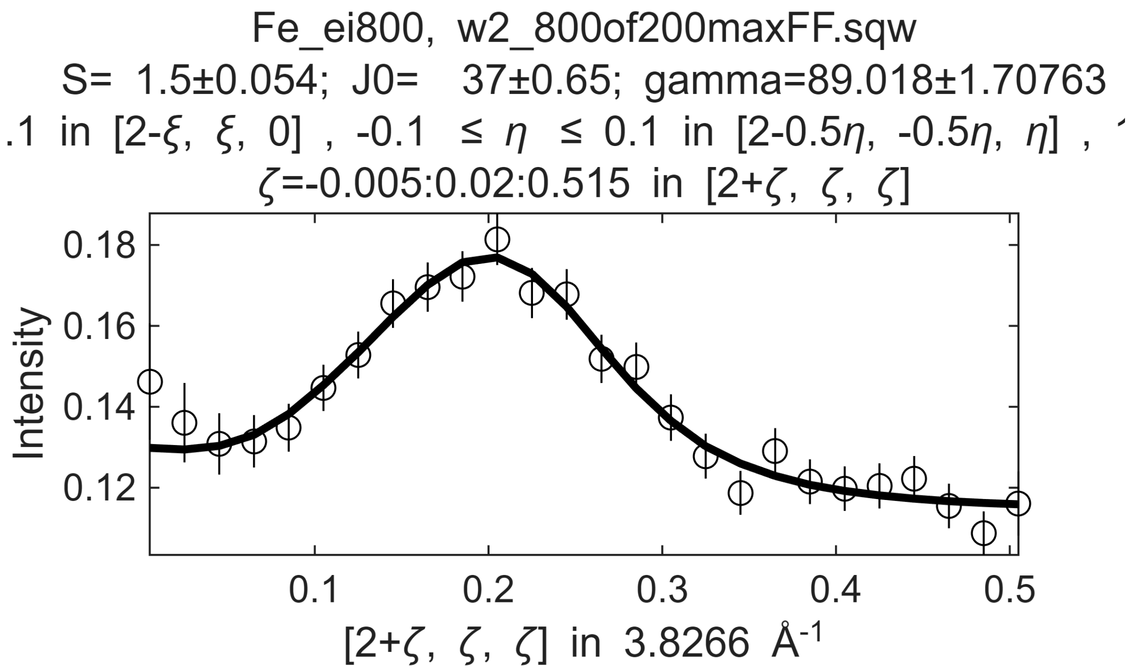

```
******************************
**** En = 135±5 going to 365
******************************
```

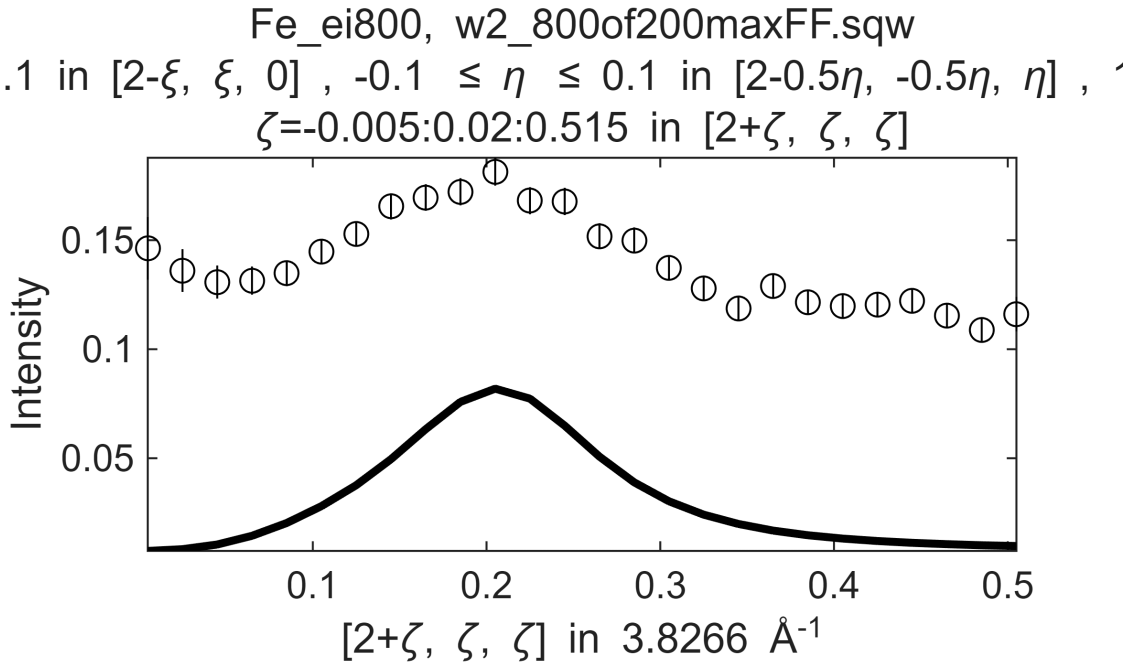
Fe_ei800, w2_800of200maxFF.sqw
S= 1.6±0.55; J0= 39± 3.9; gamma=99.2391±41.8138
).1 ≤ ξ ≤ 0.1 in [2-ξ, ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2-0.5η, -0.5η, η] , 130 ≤ E ≤
ζ=-0.005:0.02:0.515 in [2+ζ, ζ, ζ]



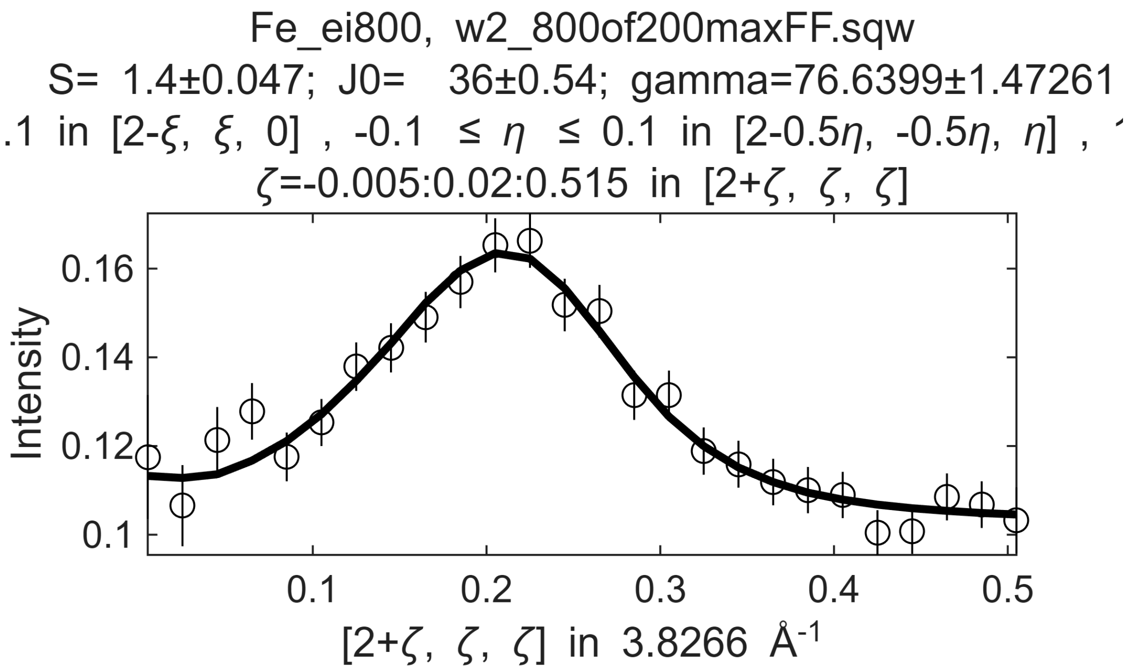


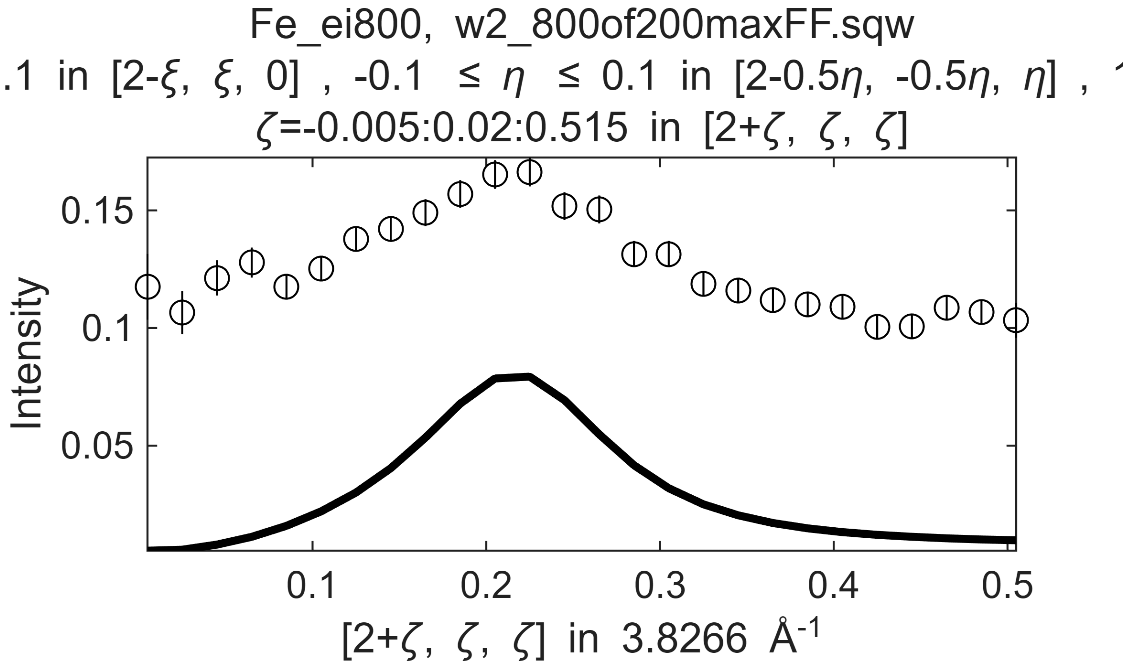

```
******************************
**** En = 145±5 going to 365
******************************
```



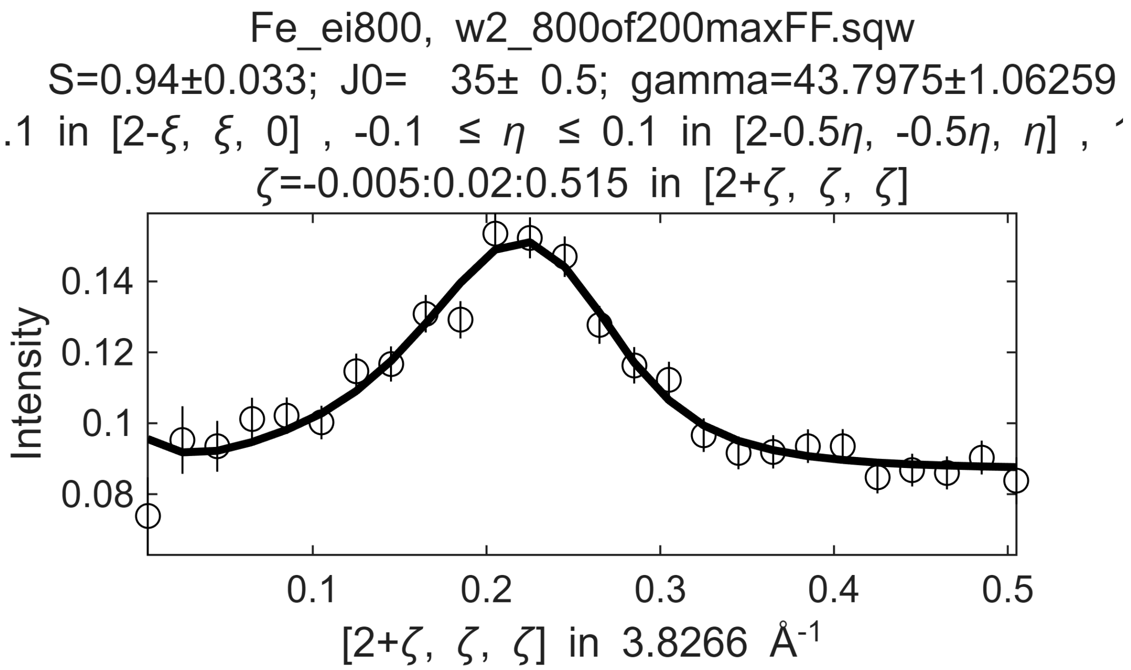


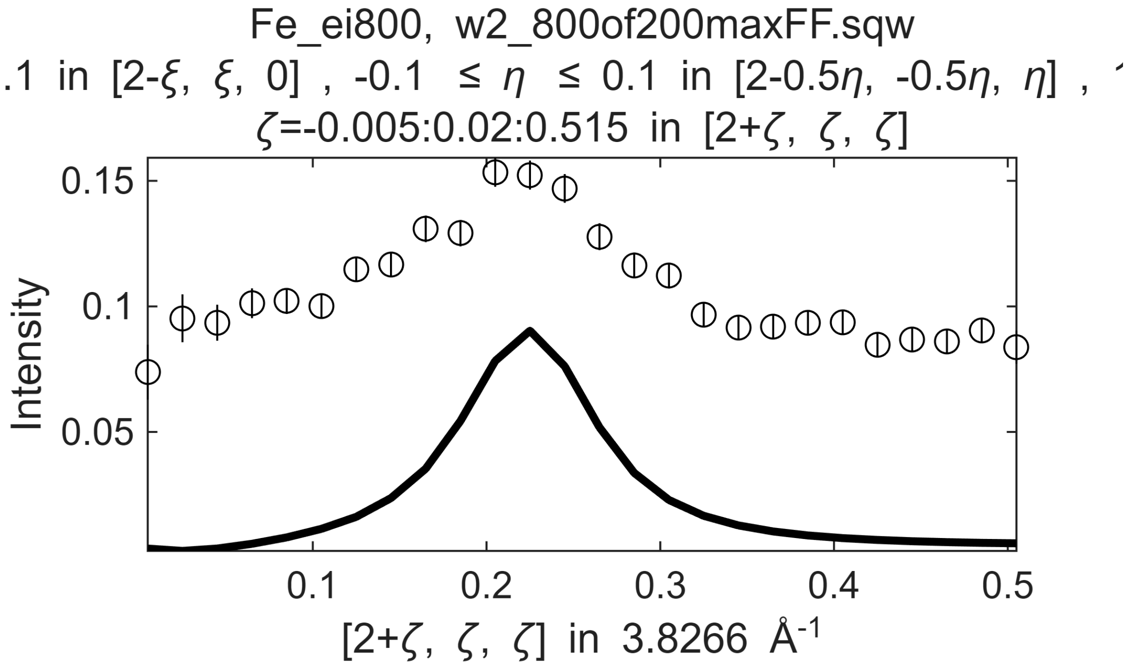

```
******************************
**** En = 155±5 going to 365
******************************
```



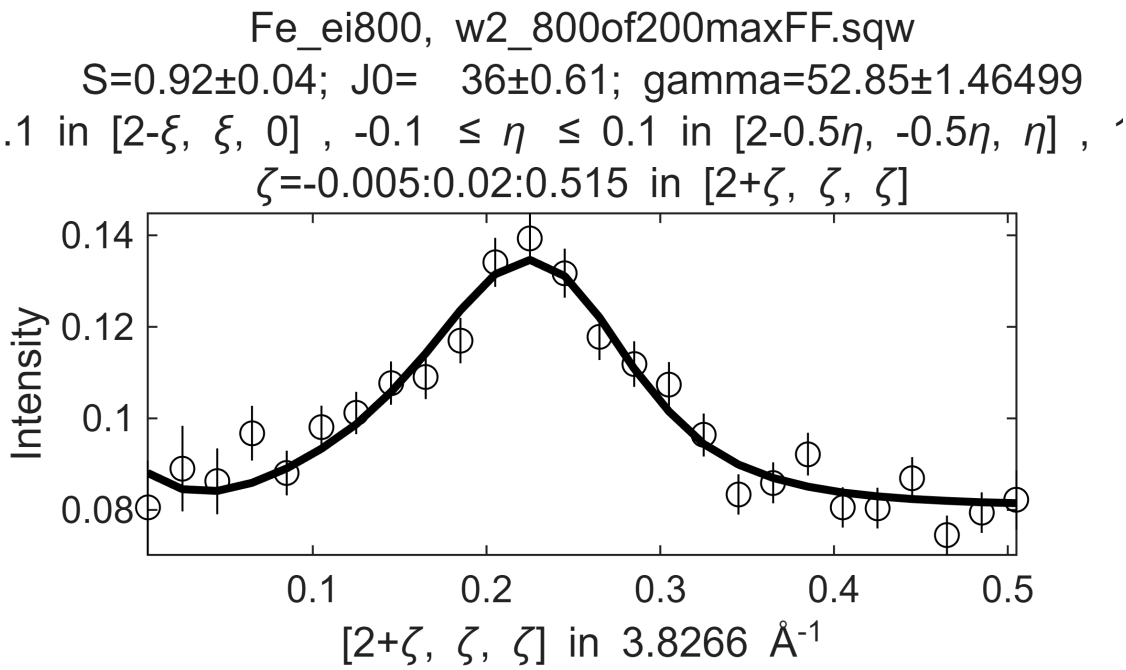


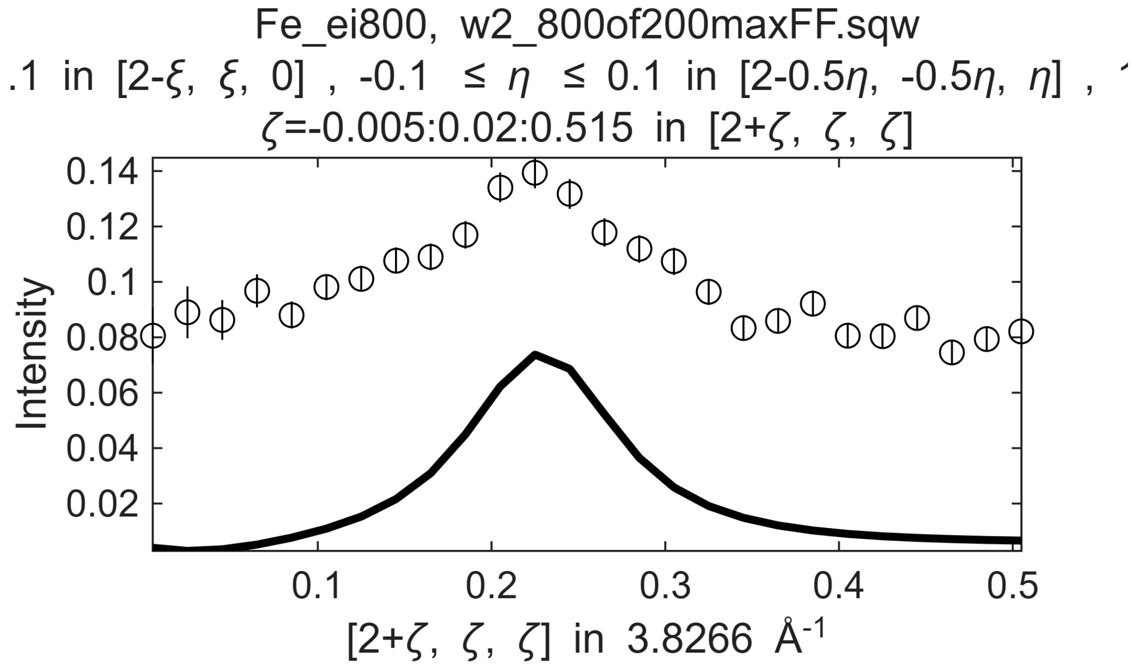

```
******************************
**** En = 165±5 going to 365
******************************
```



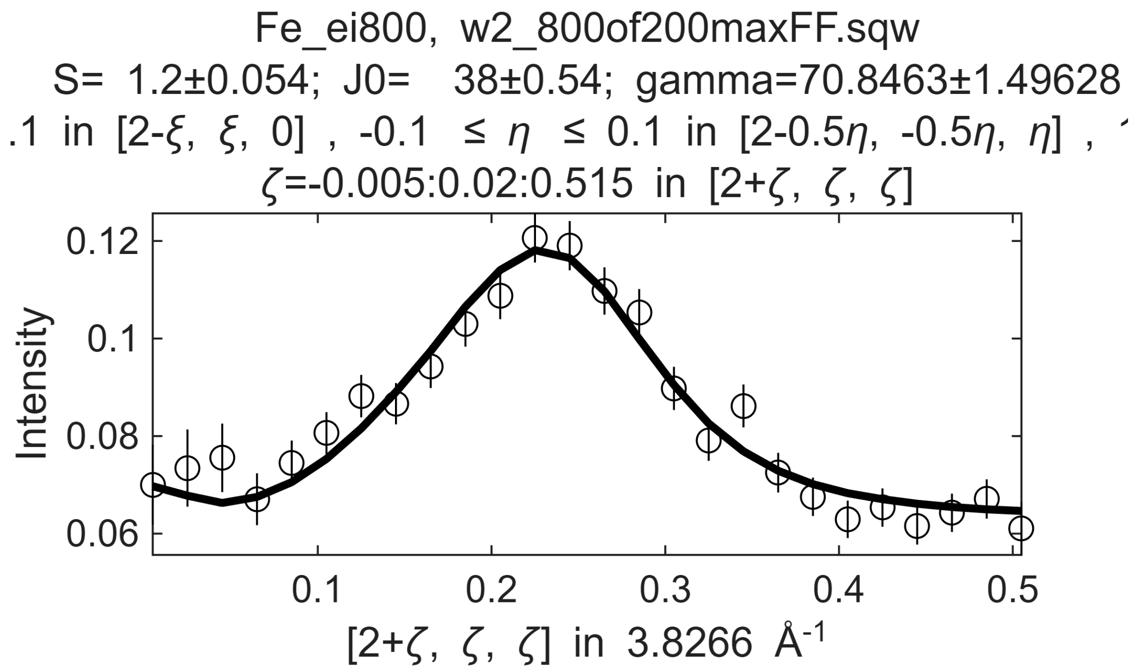


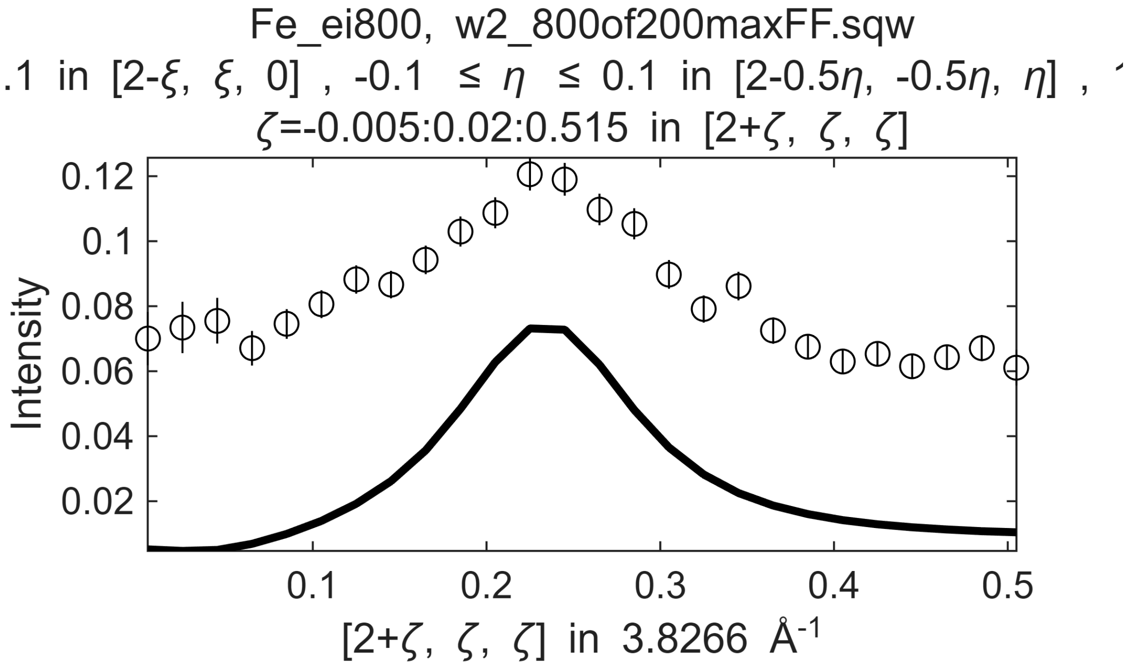
****
**** En = 175±5 going to 365
****



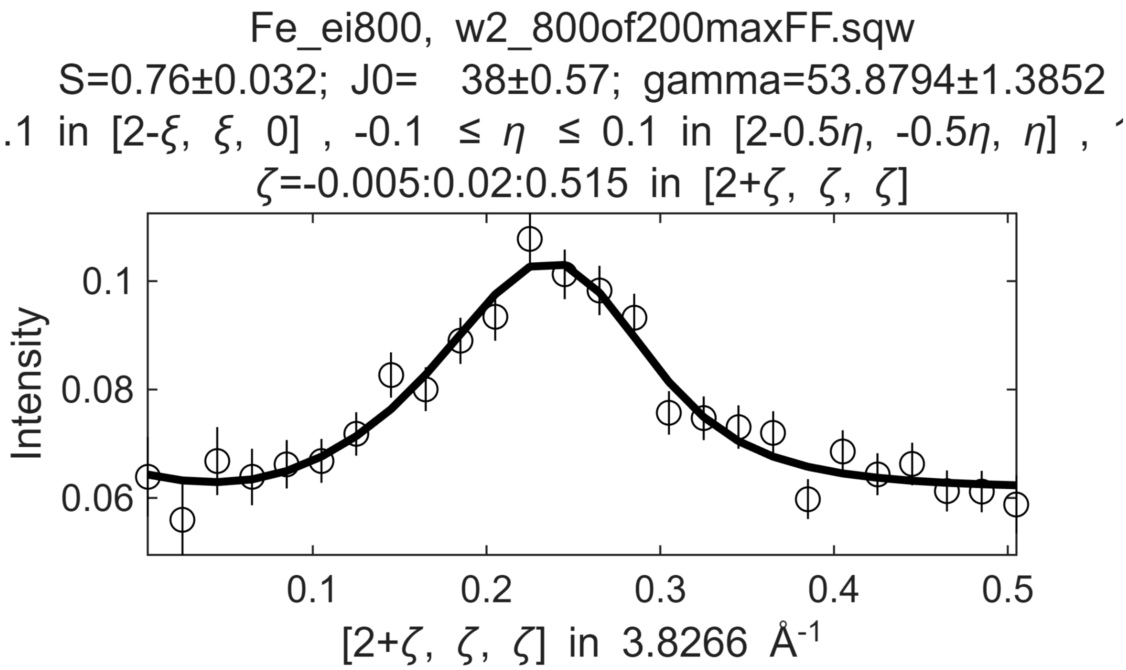


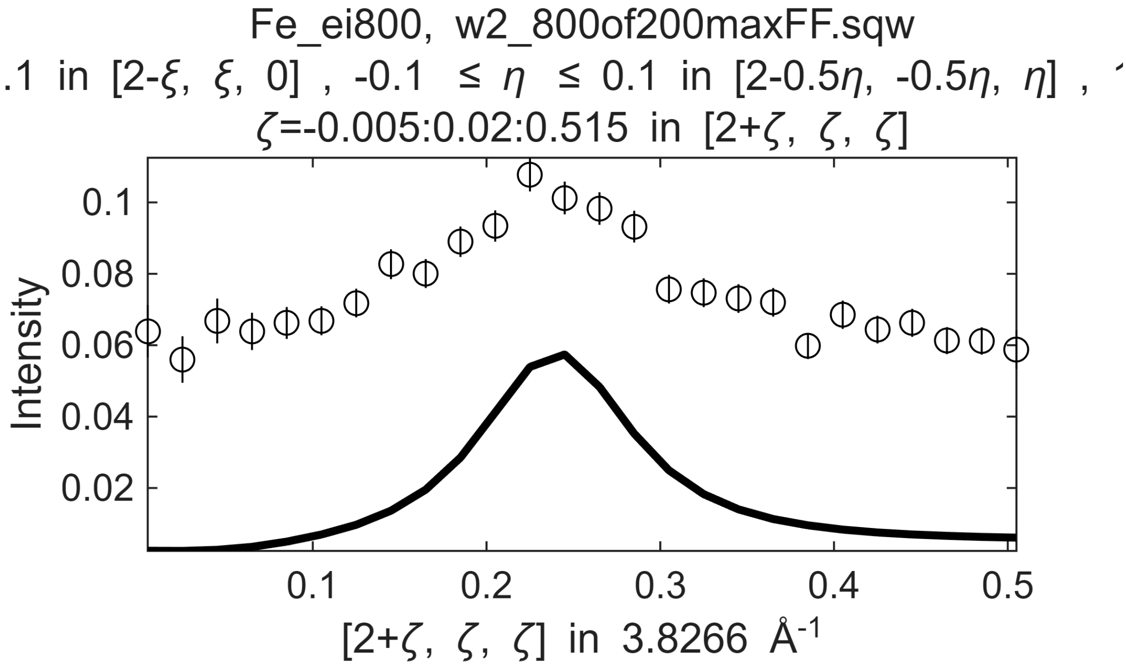

```
******************************
**** En = 185±5 going to 365
******************************
```



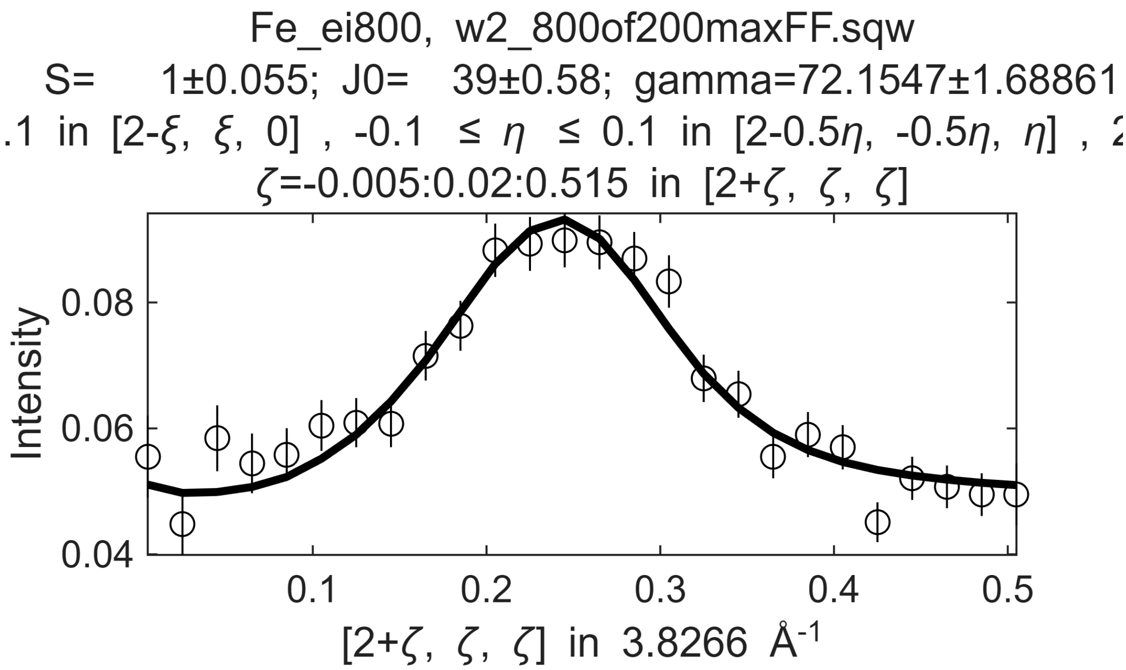


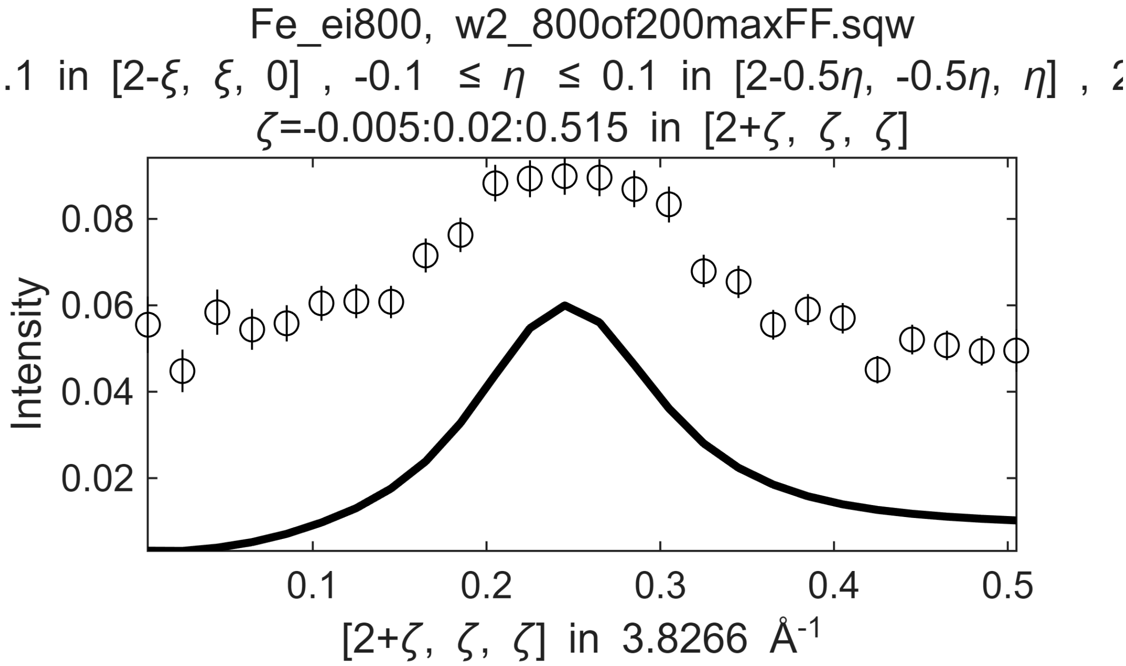

```
******************************
**** En = 195±5 going to 365
******************************
```



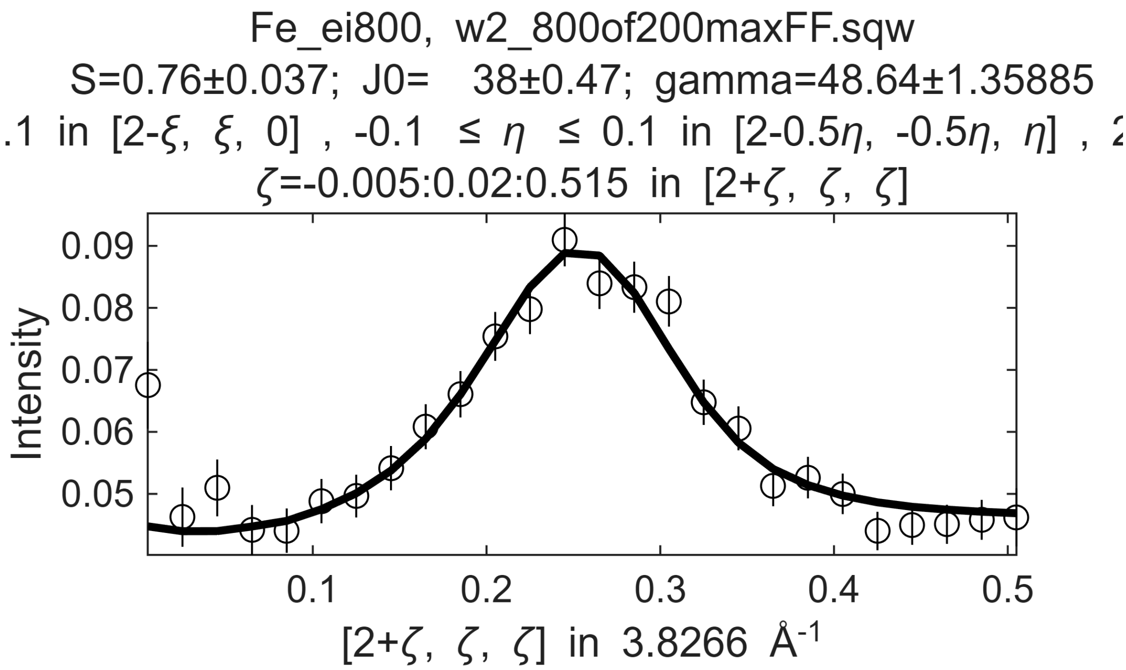


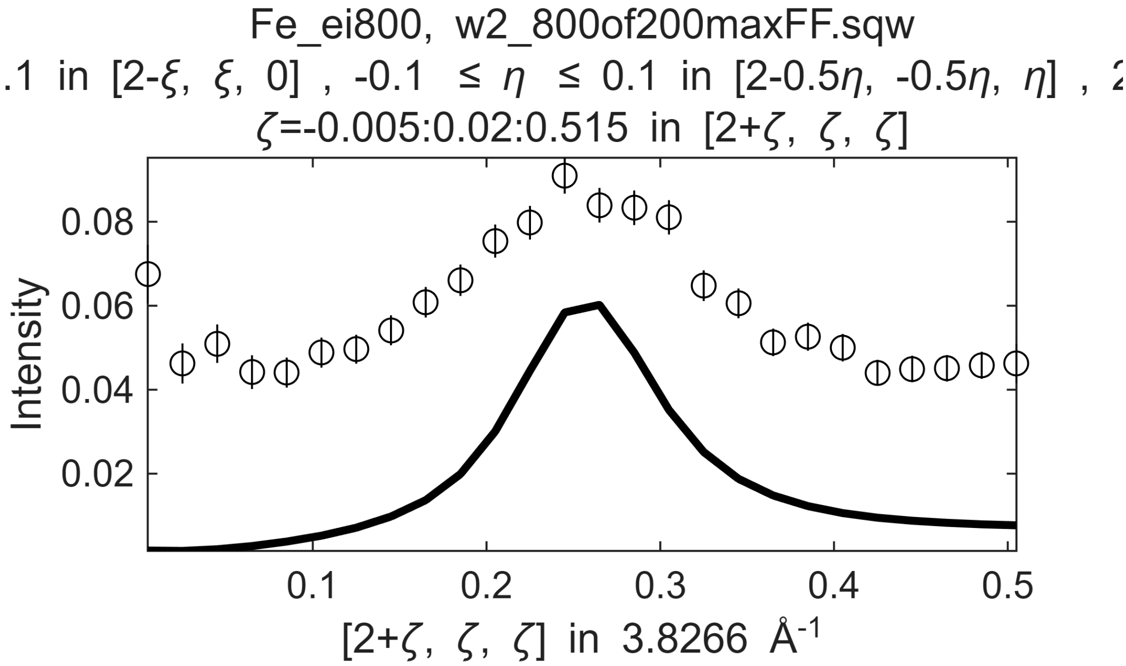

```
******************************
**** En = 205±5 going to 365
******************************
```



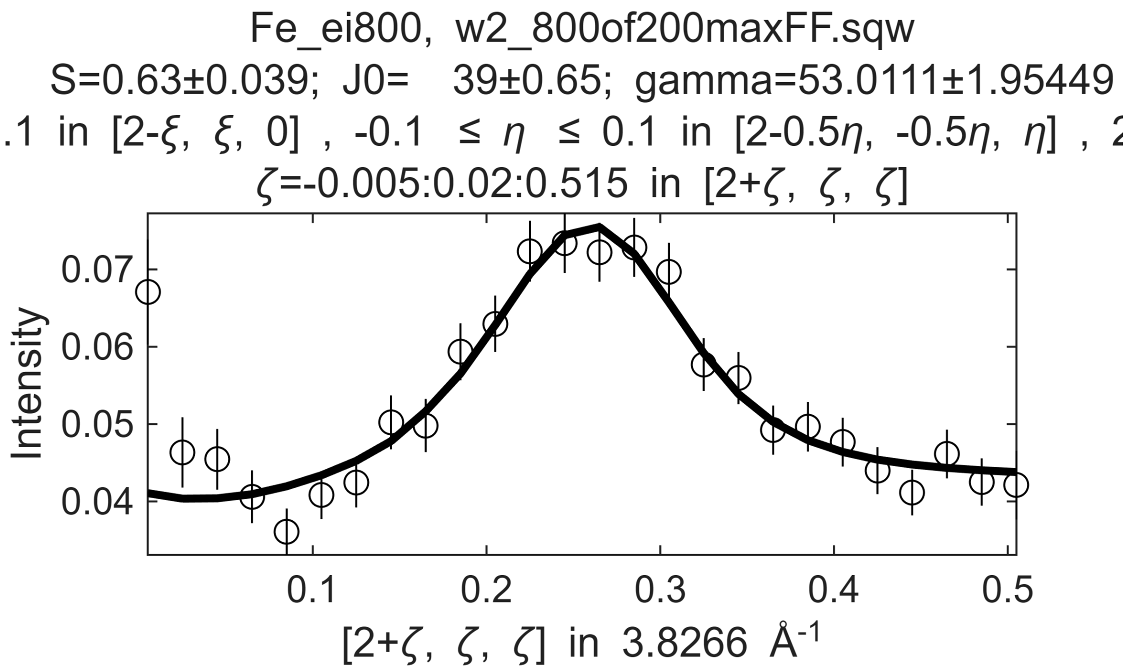


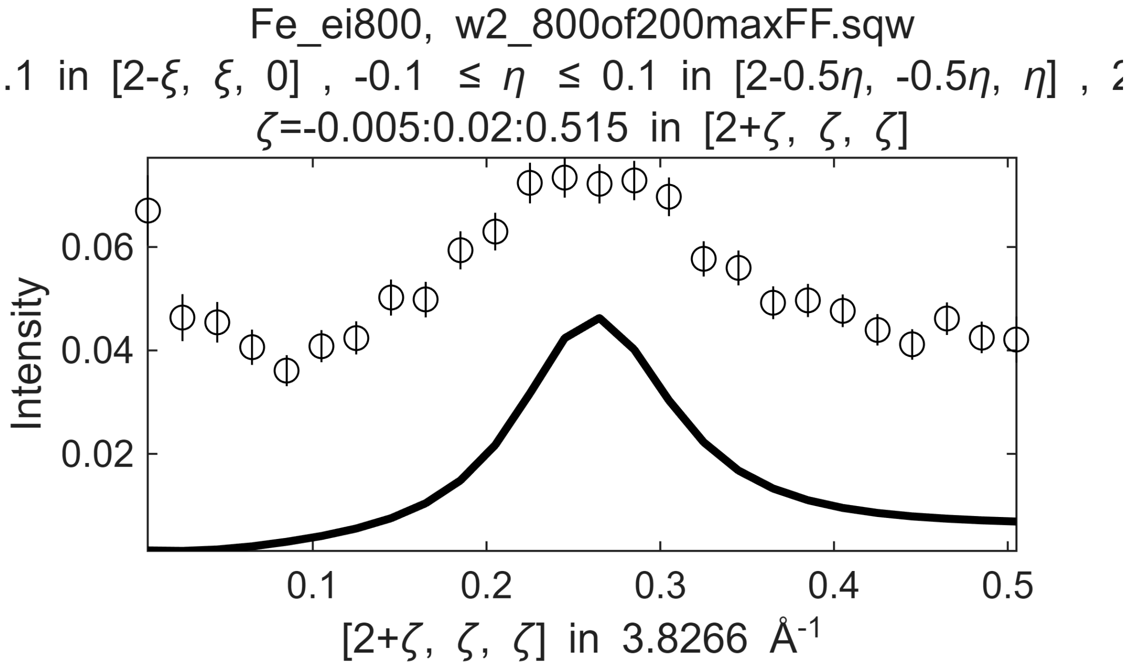

```
******************************
**** En = 215±5 going to 365
******************************
```



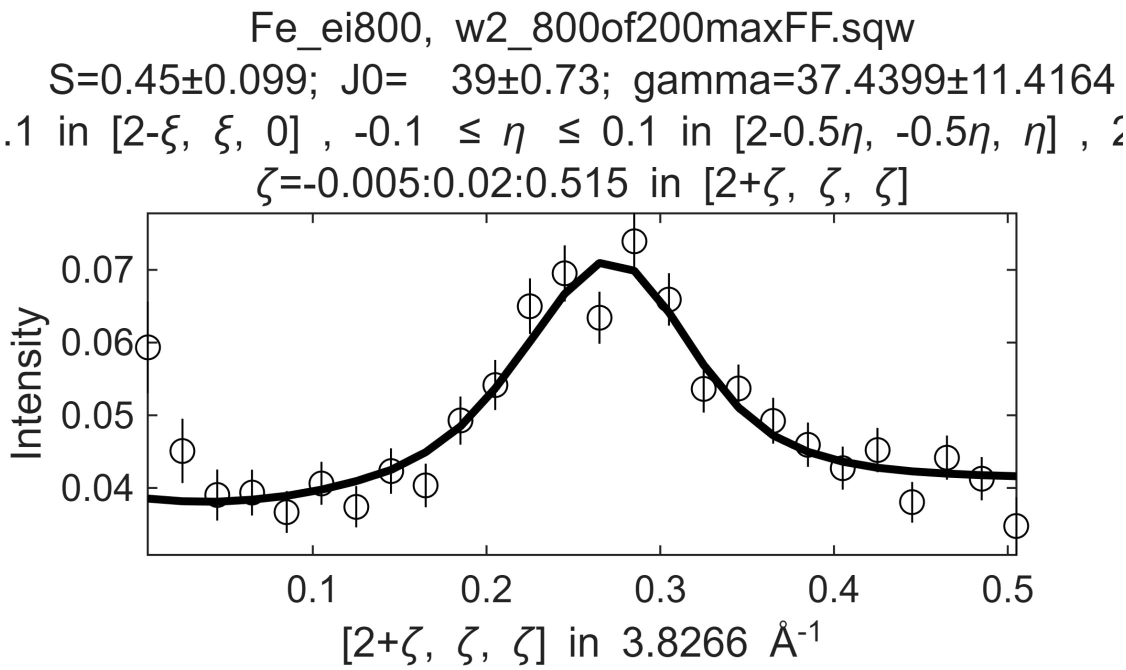


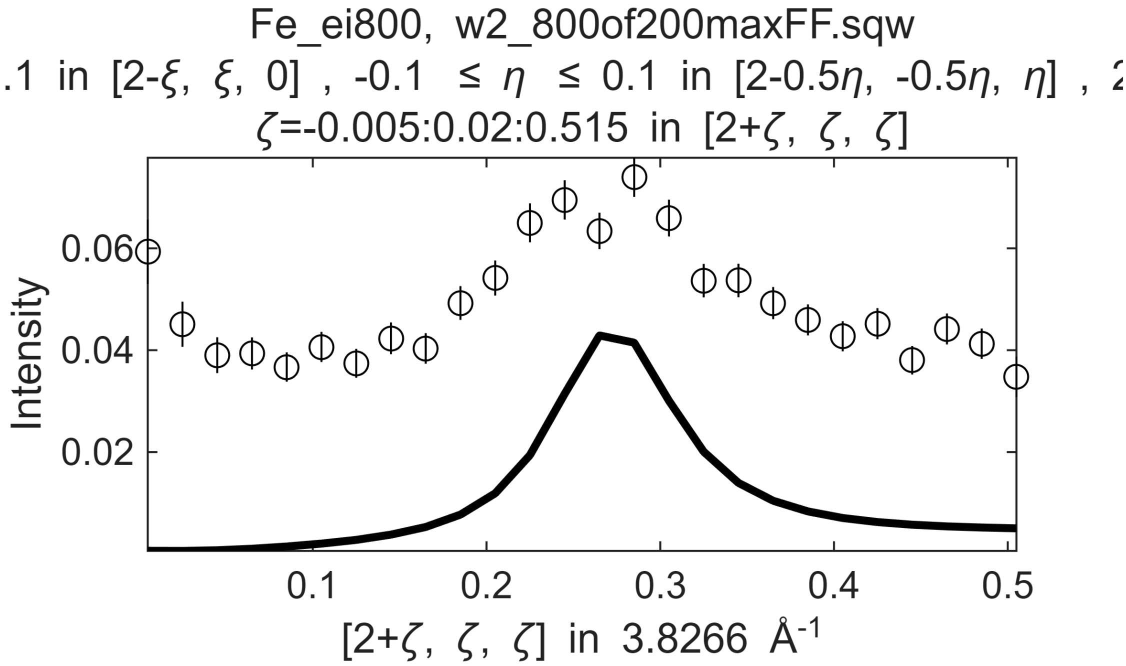

```
******************************
**** En = 225±5 going to 365
******************************
```



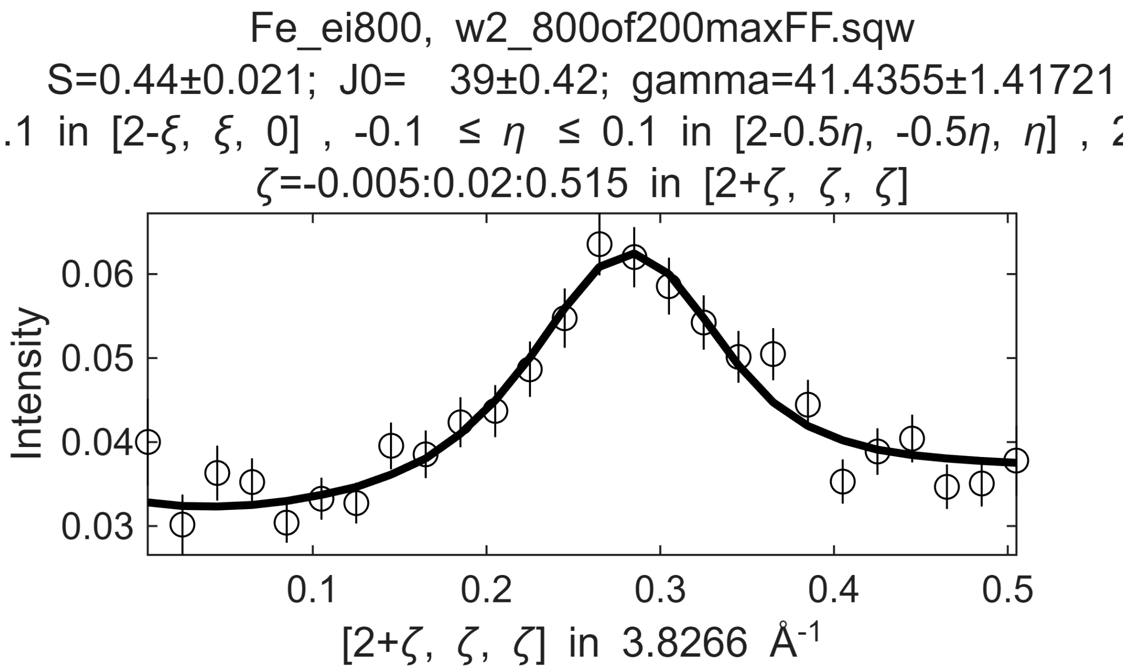


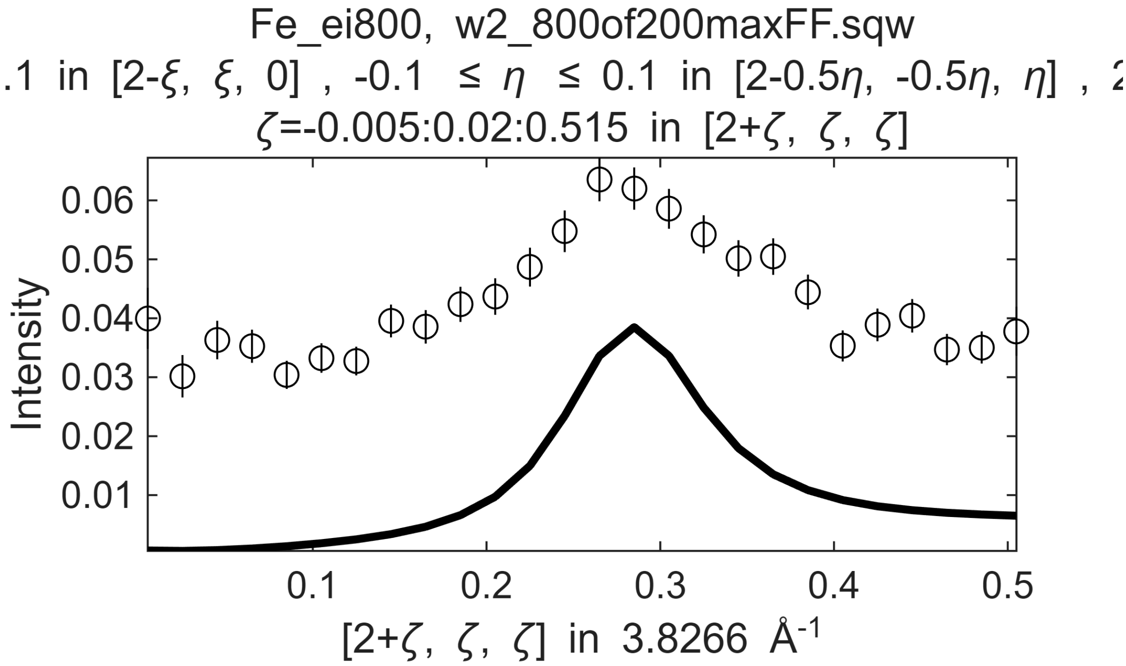

```
******************************
**** En = 235±5 going to 365
******************************
```



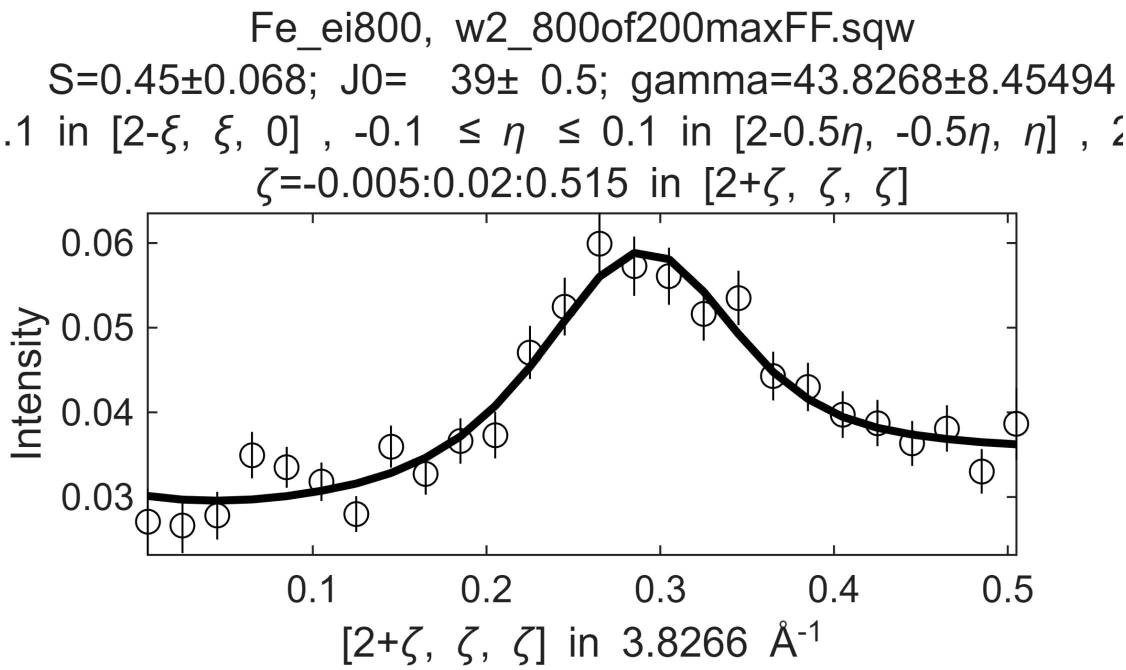


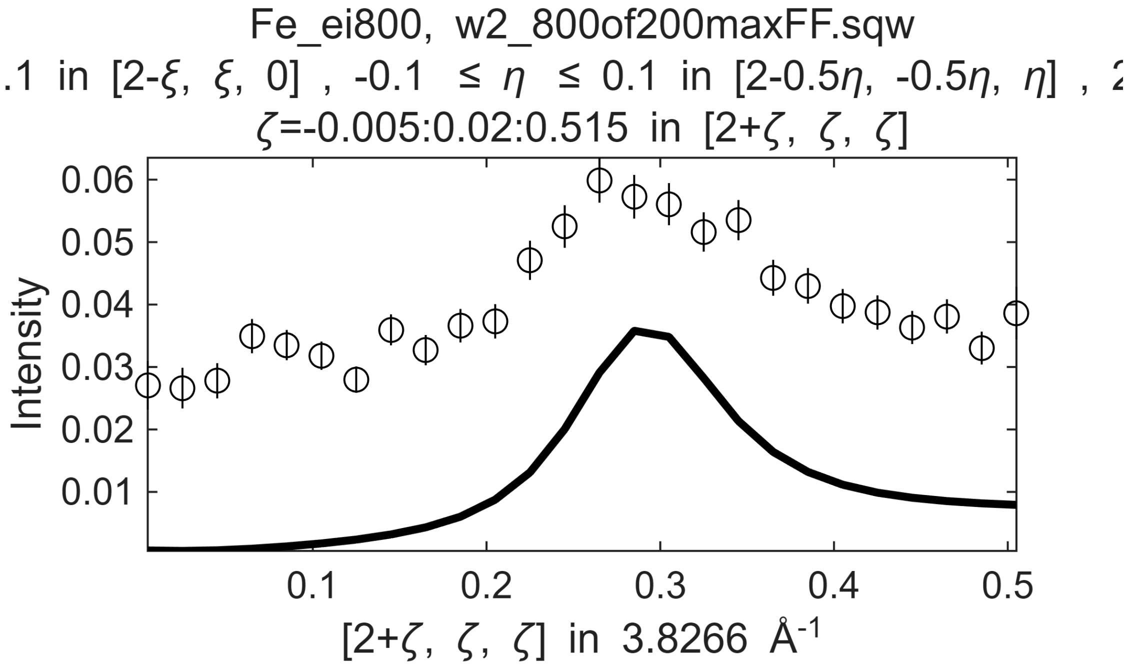

```
******************************
**** En = 245±5 going to 365
******************************
```



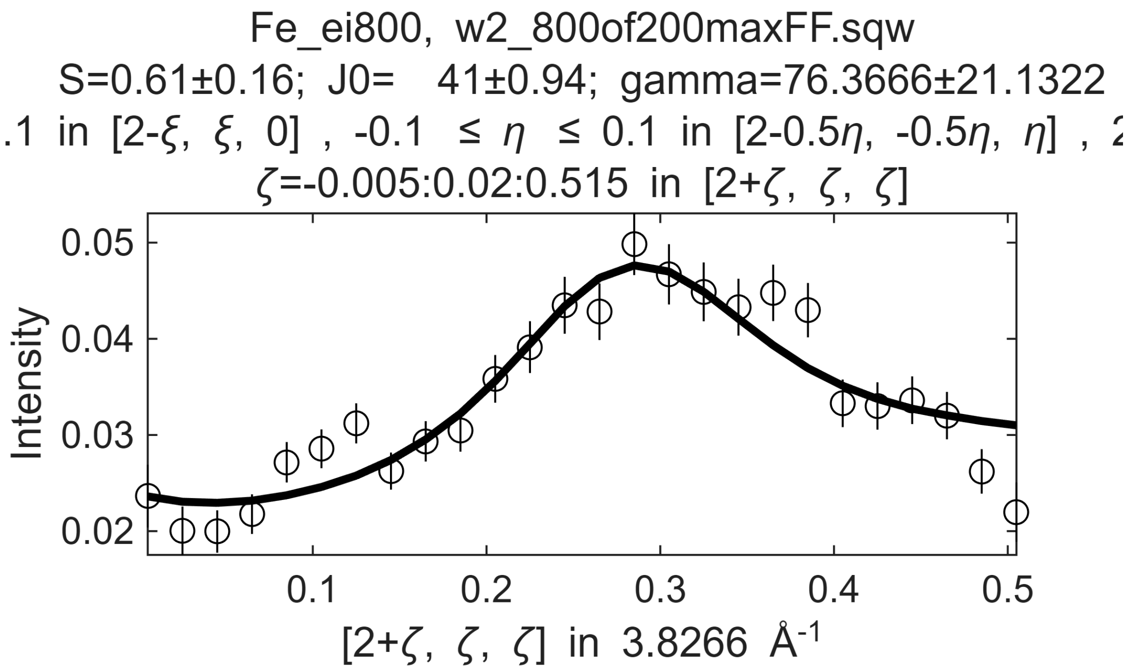


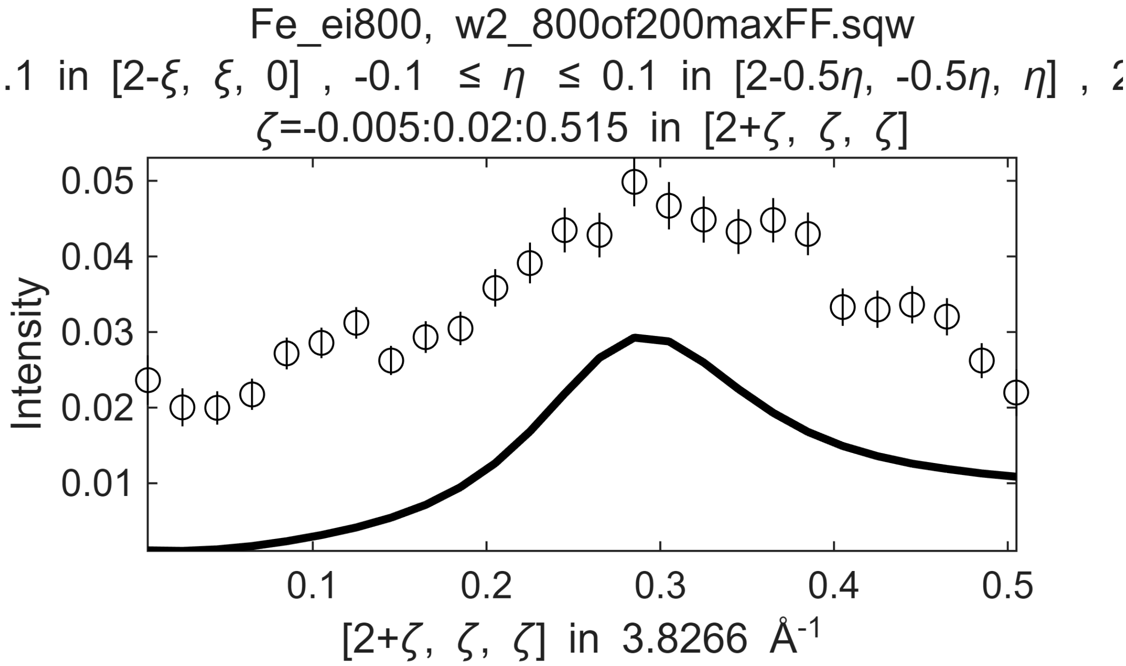
****
**** En = 255±5 going to 365
****



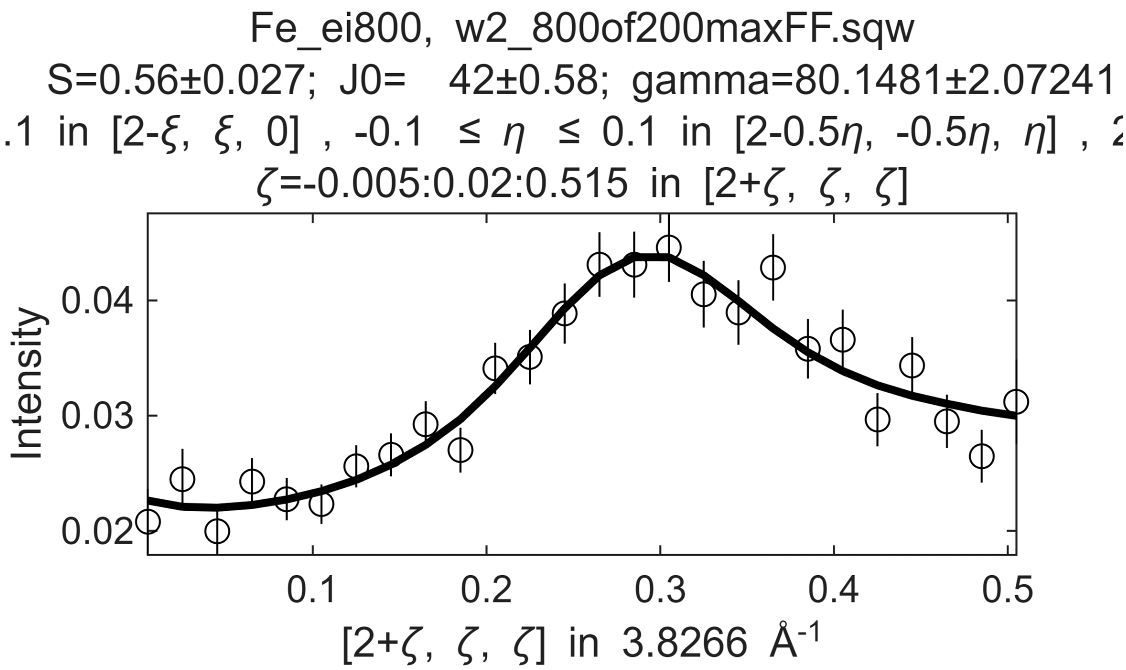


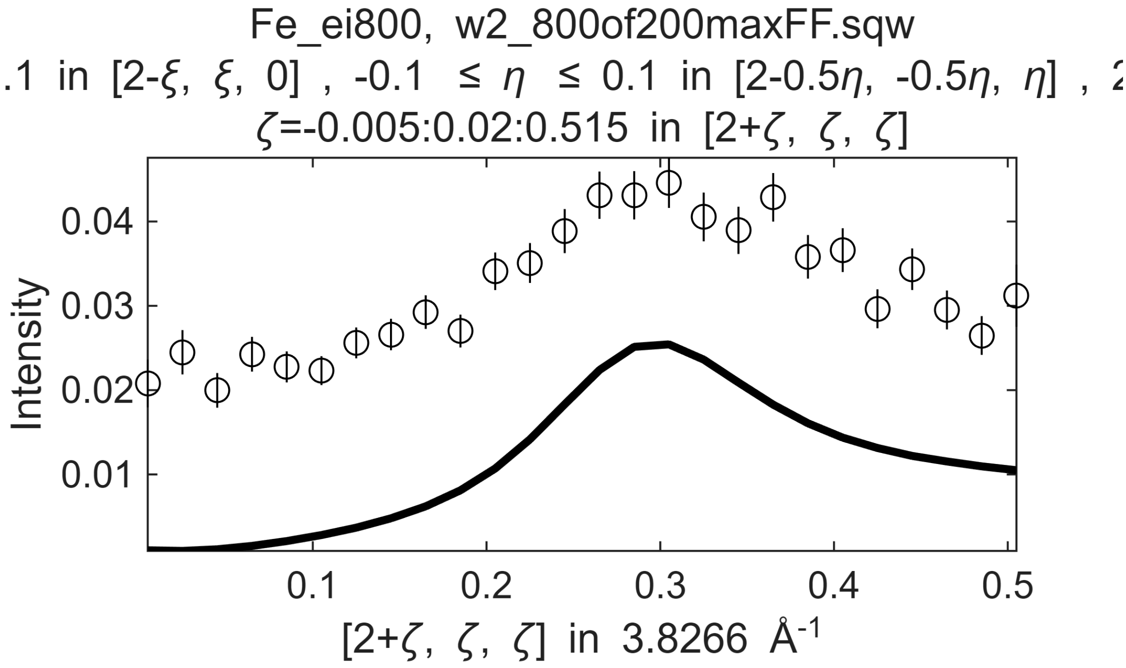

```
******************************
**** En = 265±5 going to 365
******************************
```



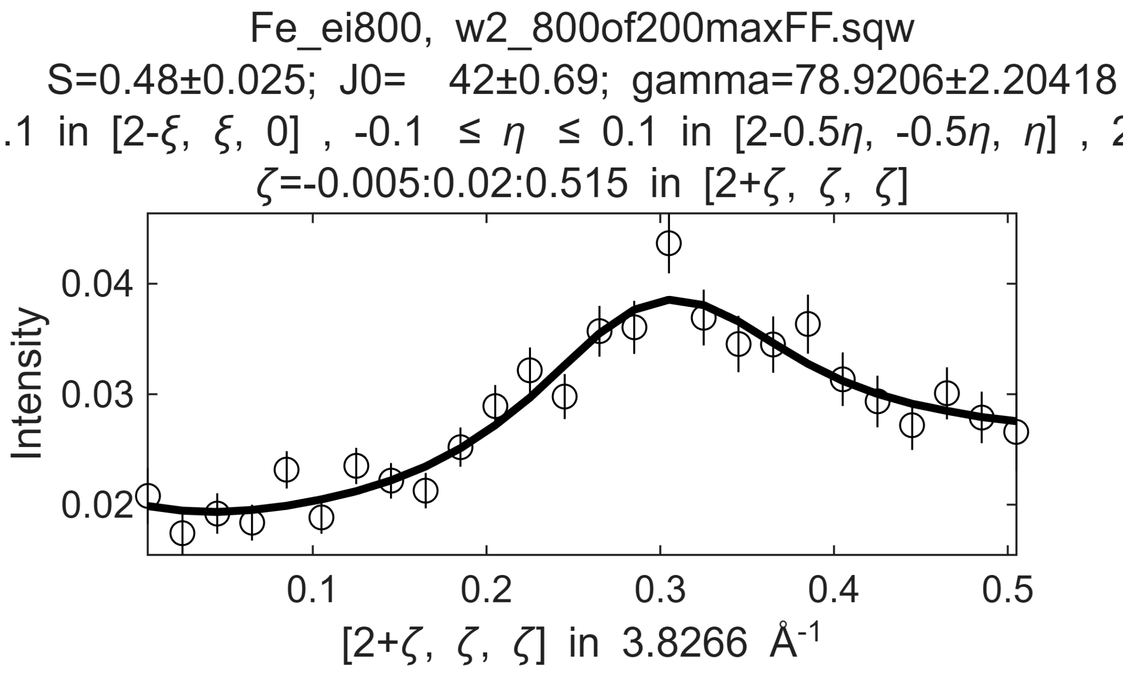


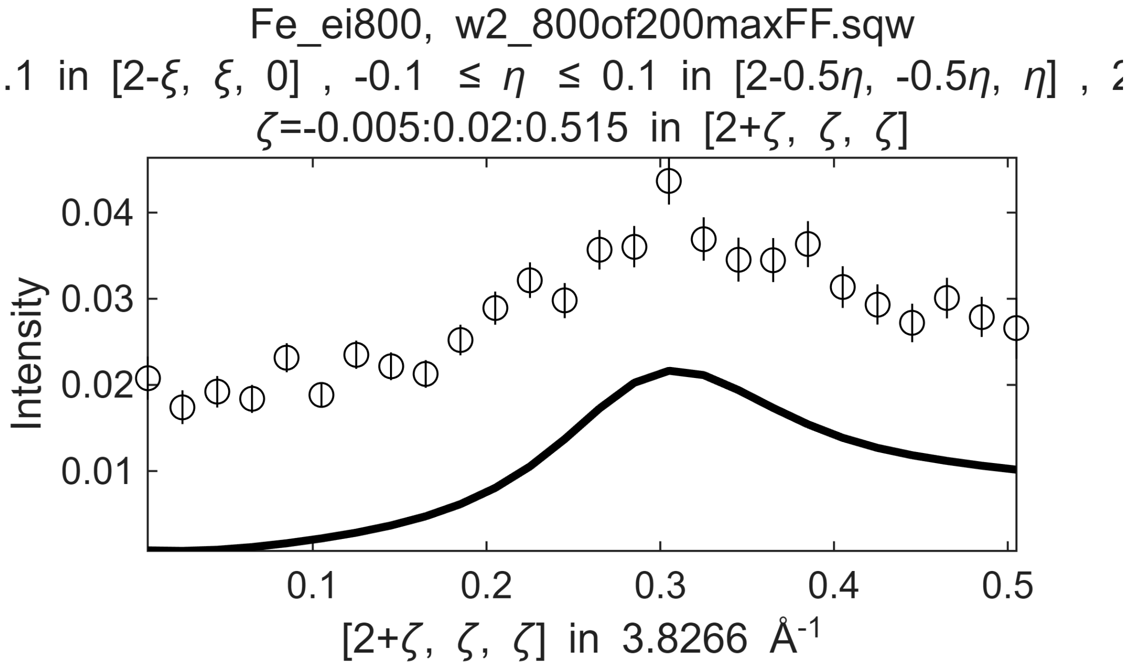

```
*****************************
**** En = 275±5 going to 365
*****************************
```



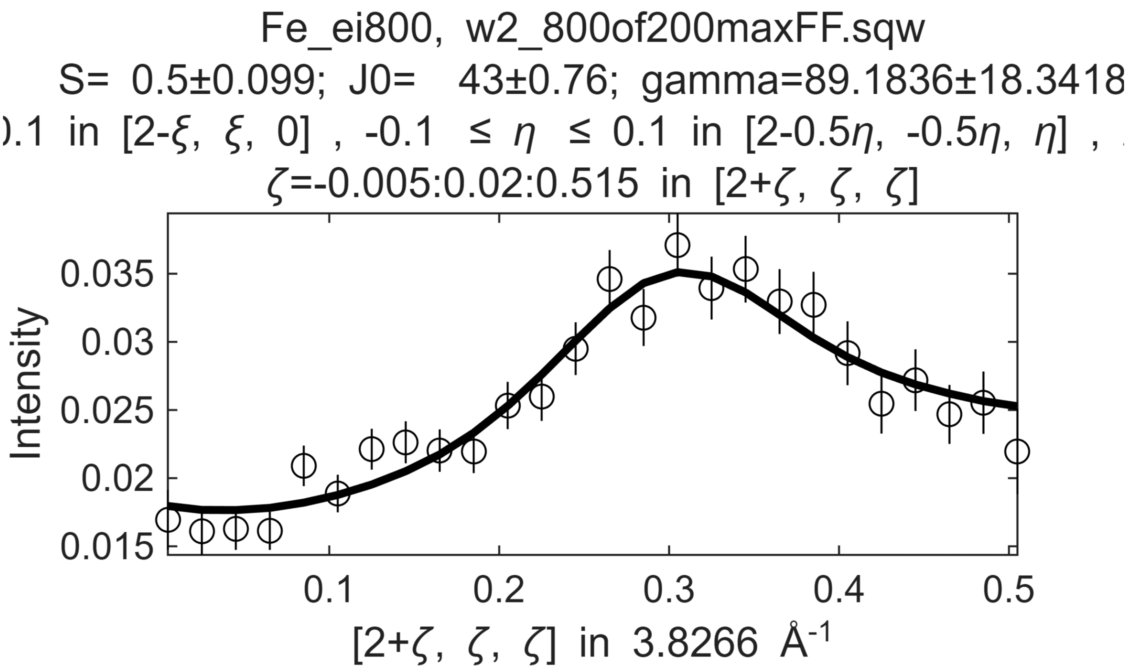


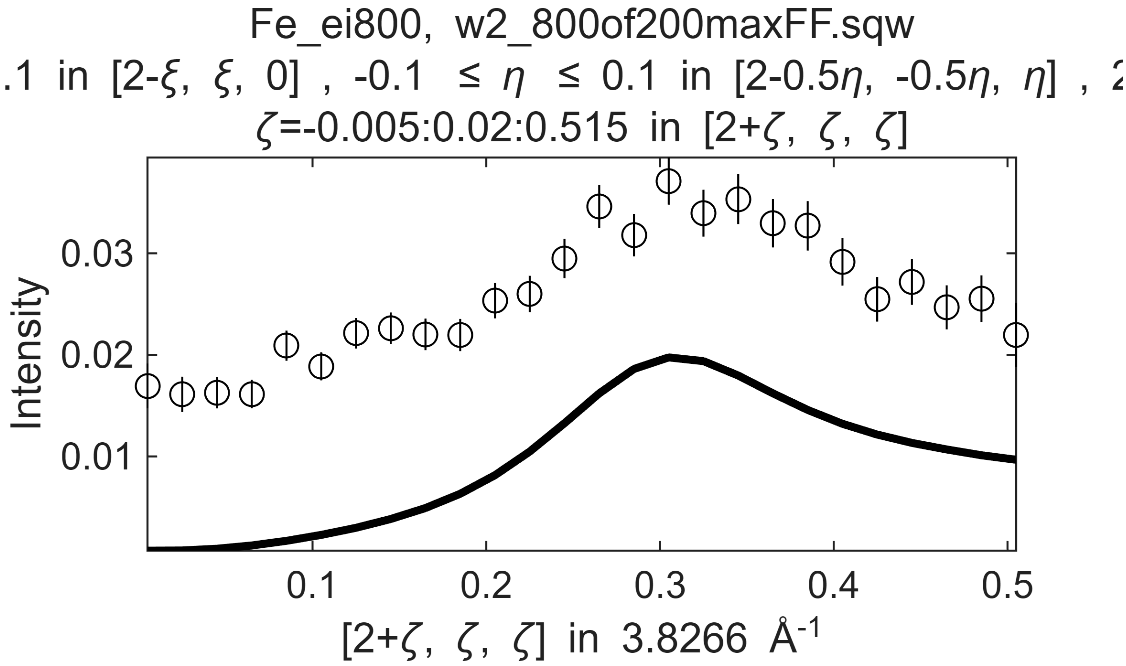
****
**** En = 285±5 going to 365
****



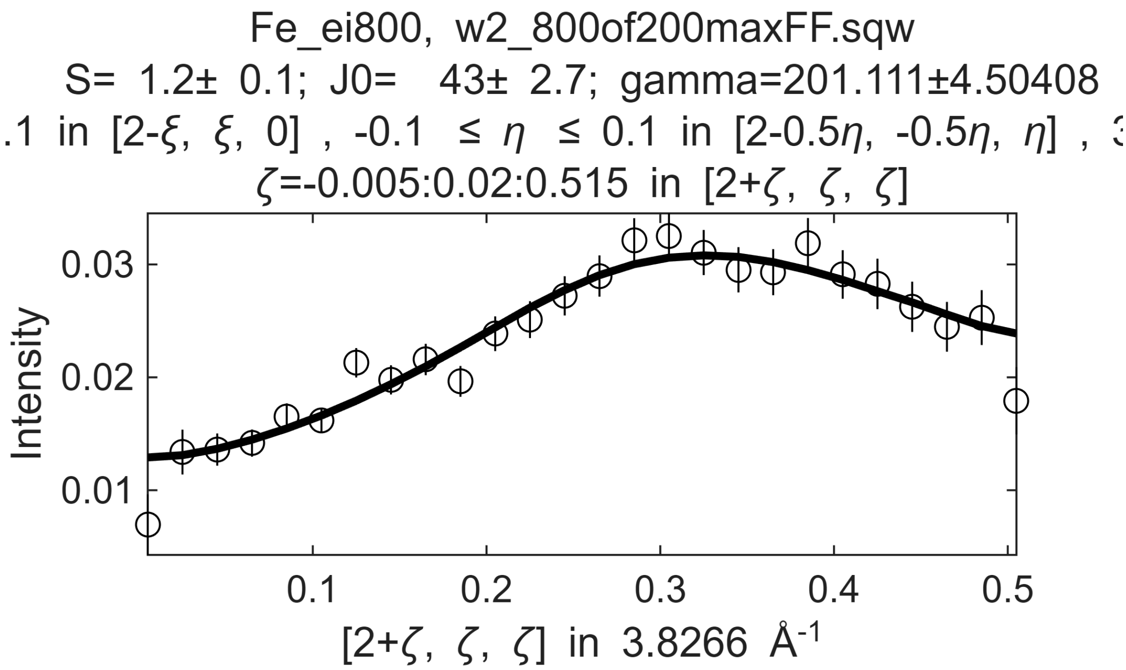


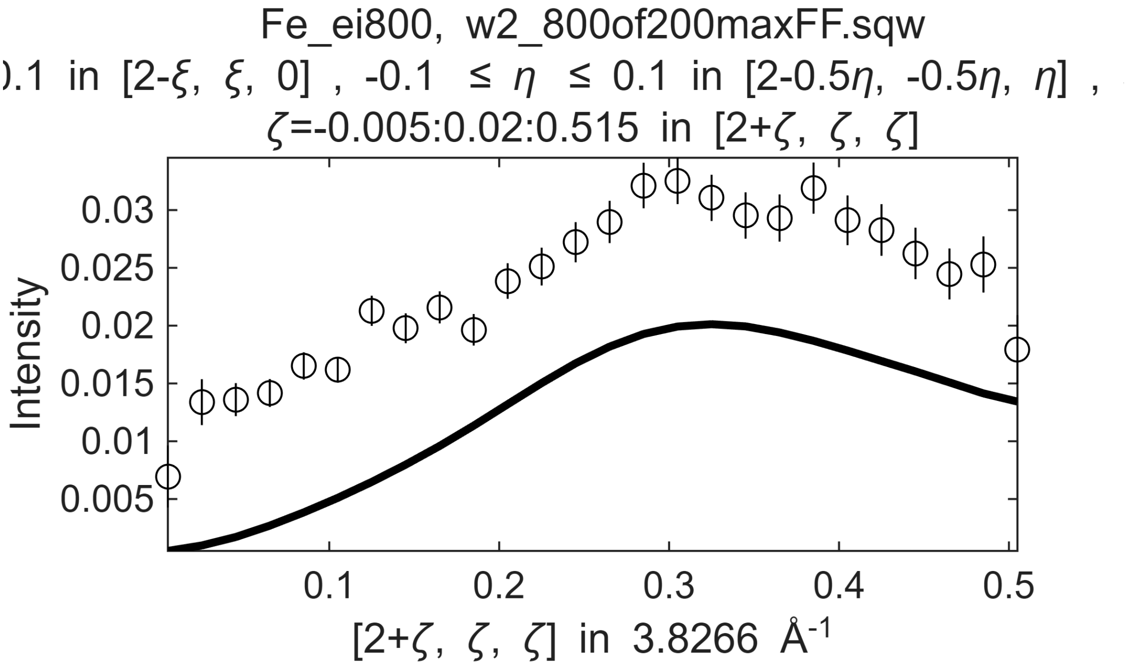

```
******************************
**** En = 295±5 going to 365
******************************
```



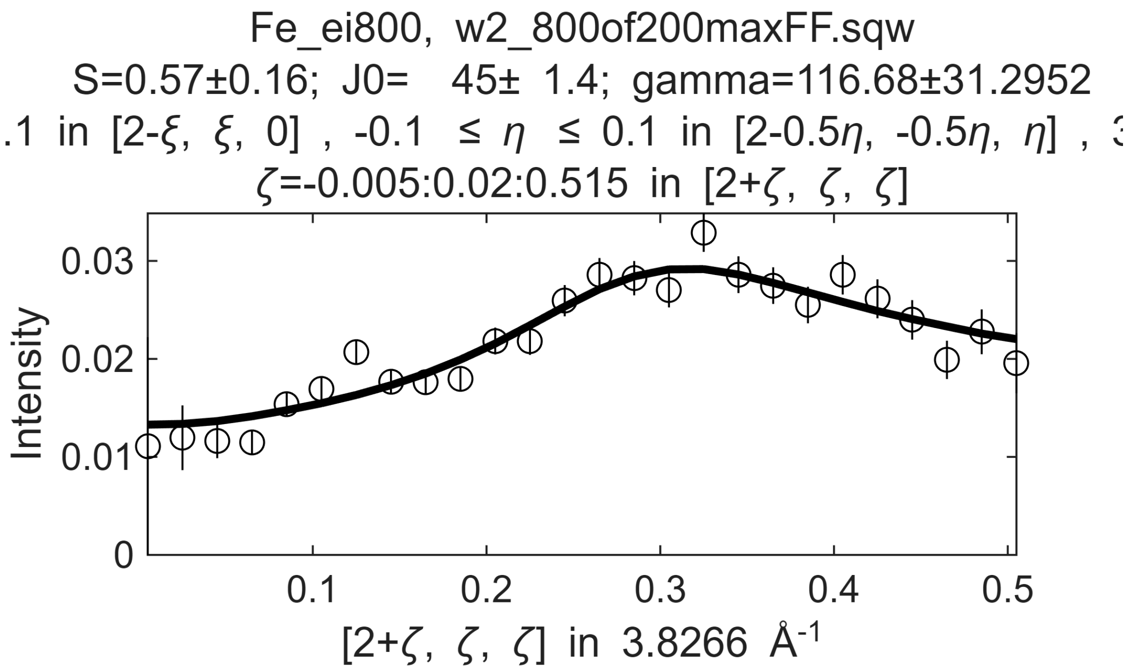


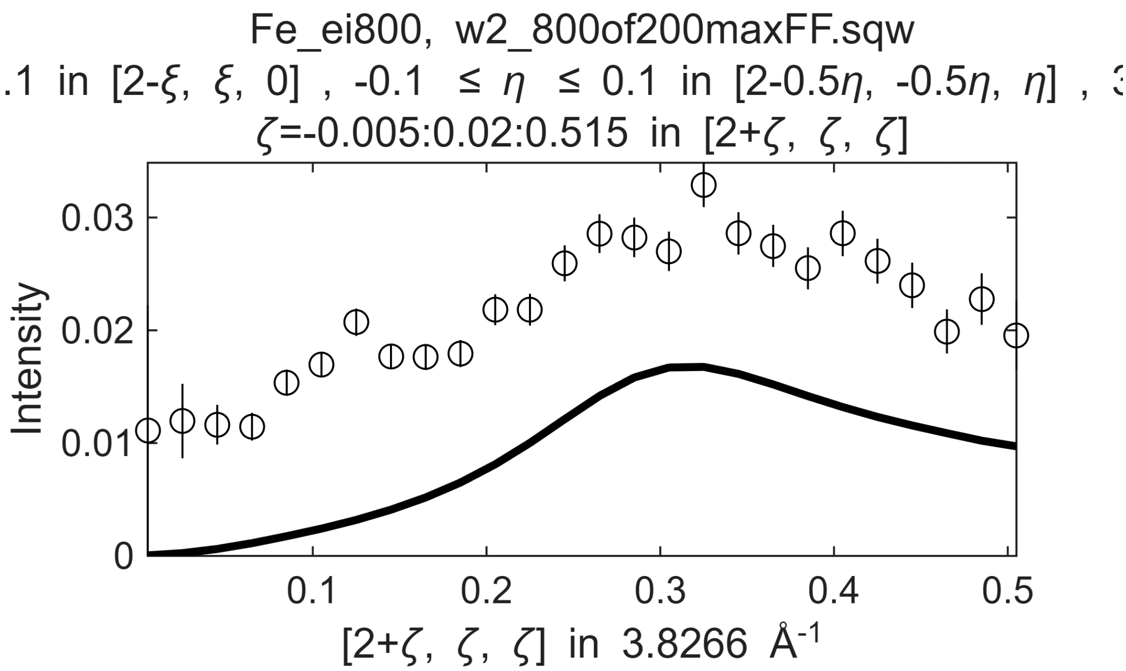

```
*****************************
**** En = 305±5 going to 365
*****************************
```



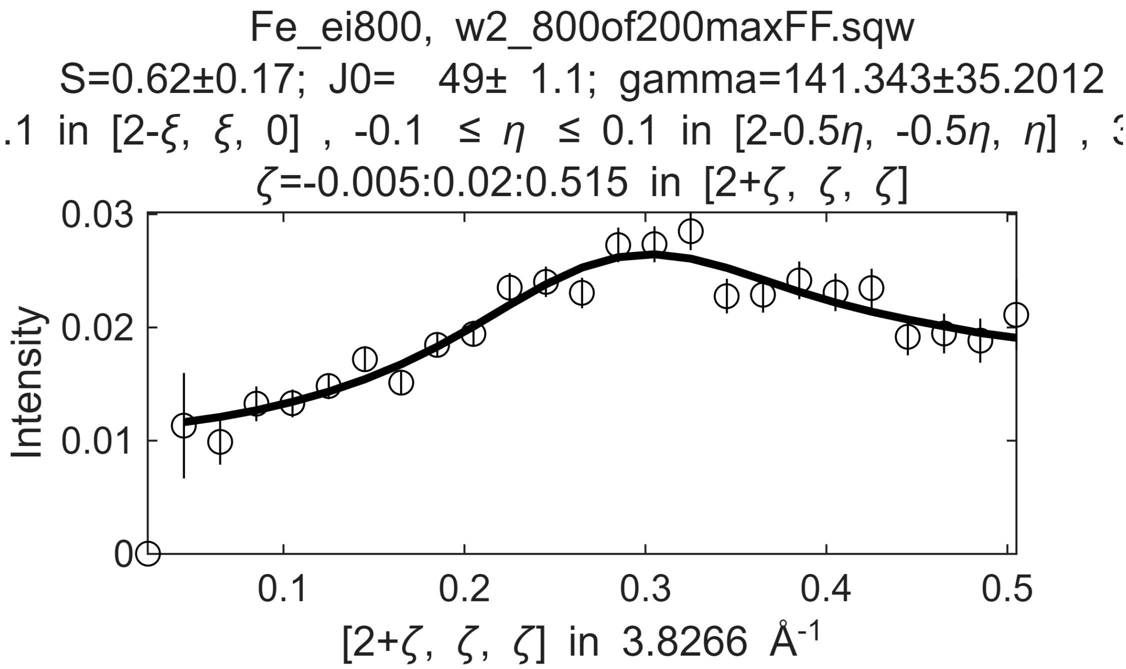


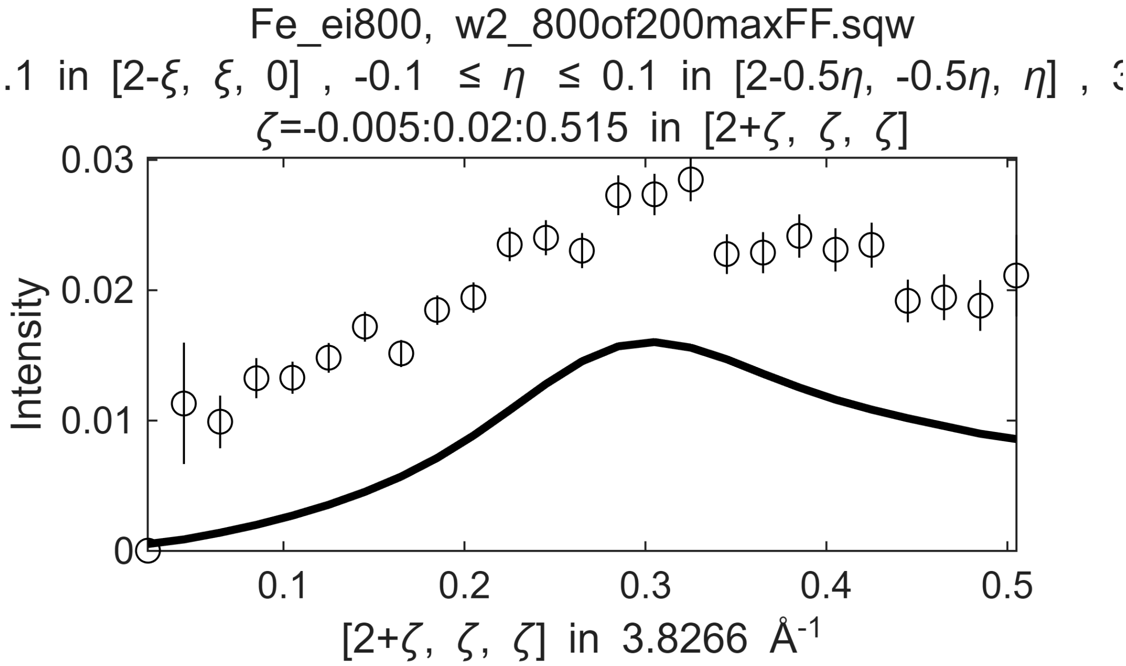

```
******************************
**** En = 315±5 going to 365
******************************
```



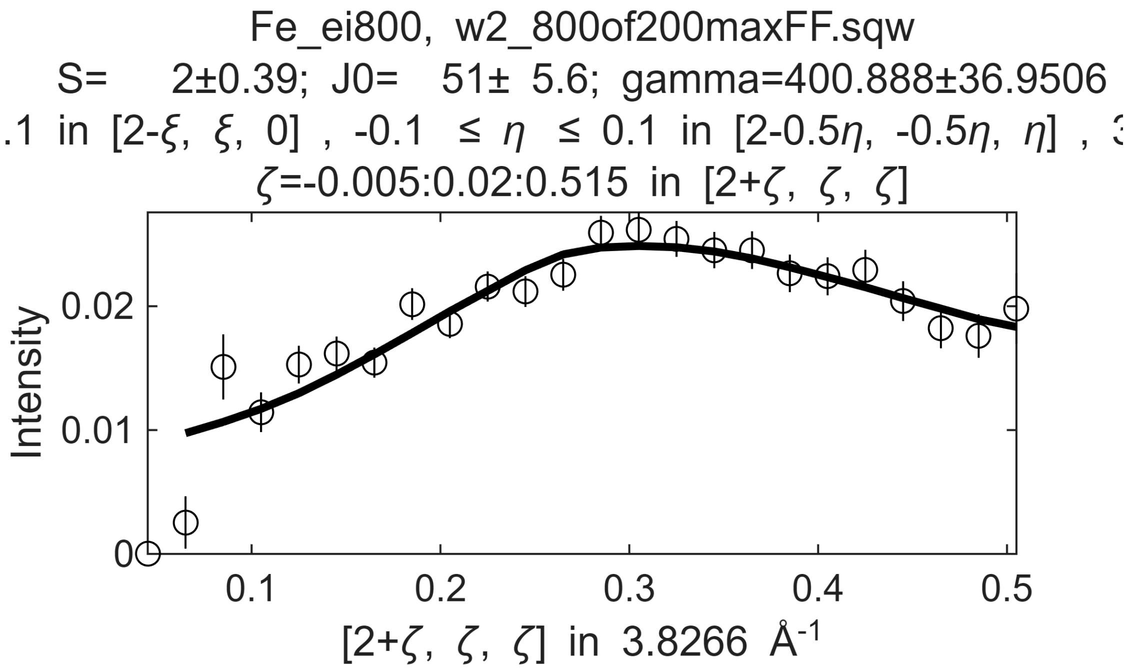


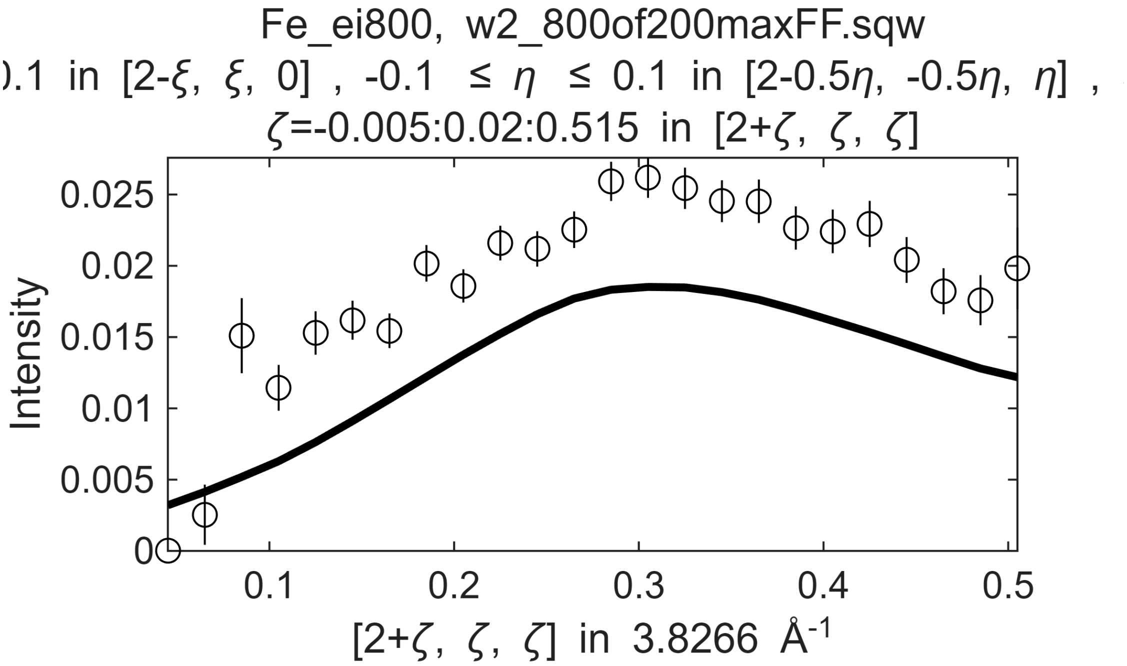

```
******************************
**** En = 325±5 going to 365
******************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 3.8462
```



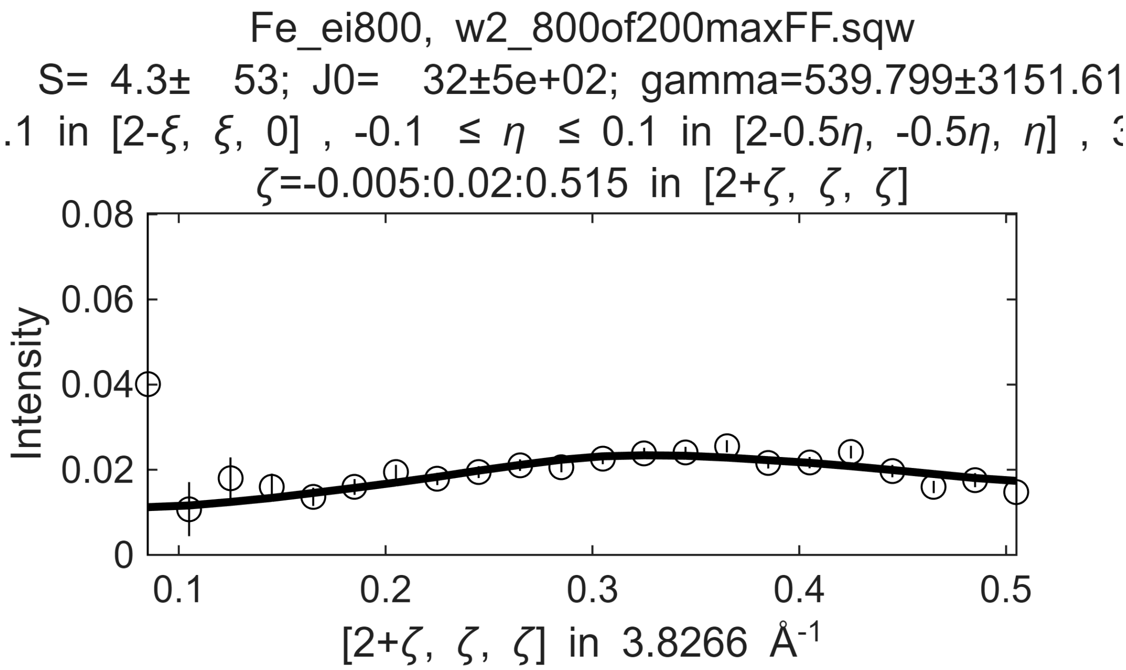


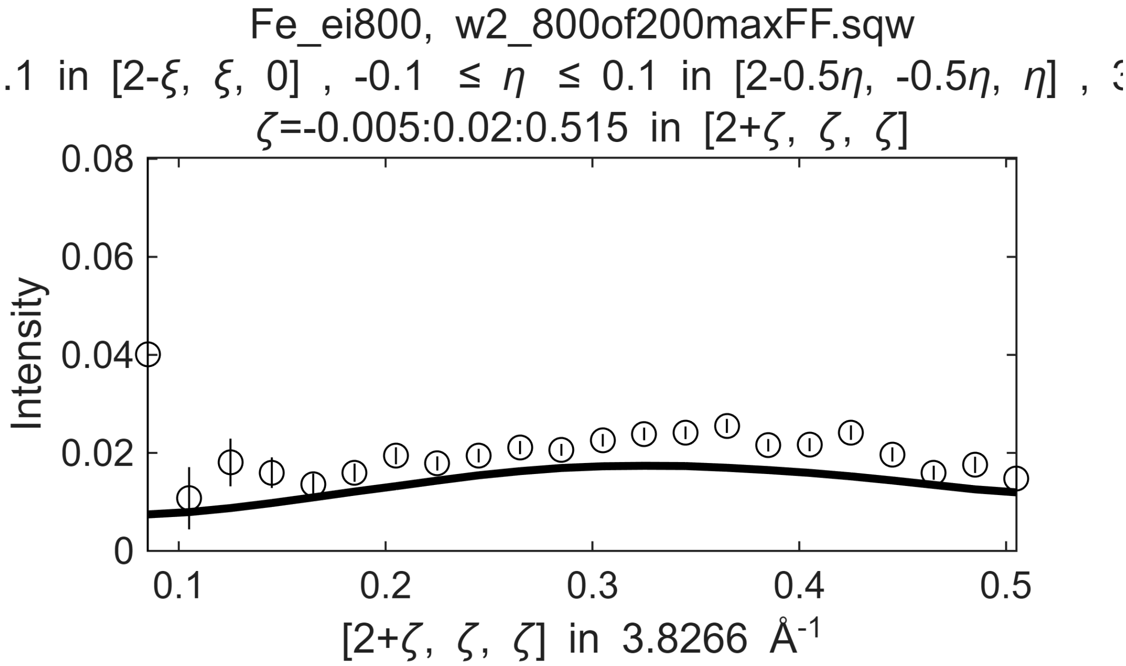

```
******************************
**** En = 335±5 going to 365
******************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 3.8462
```



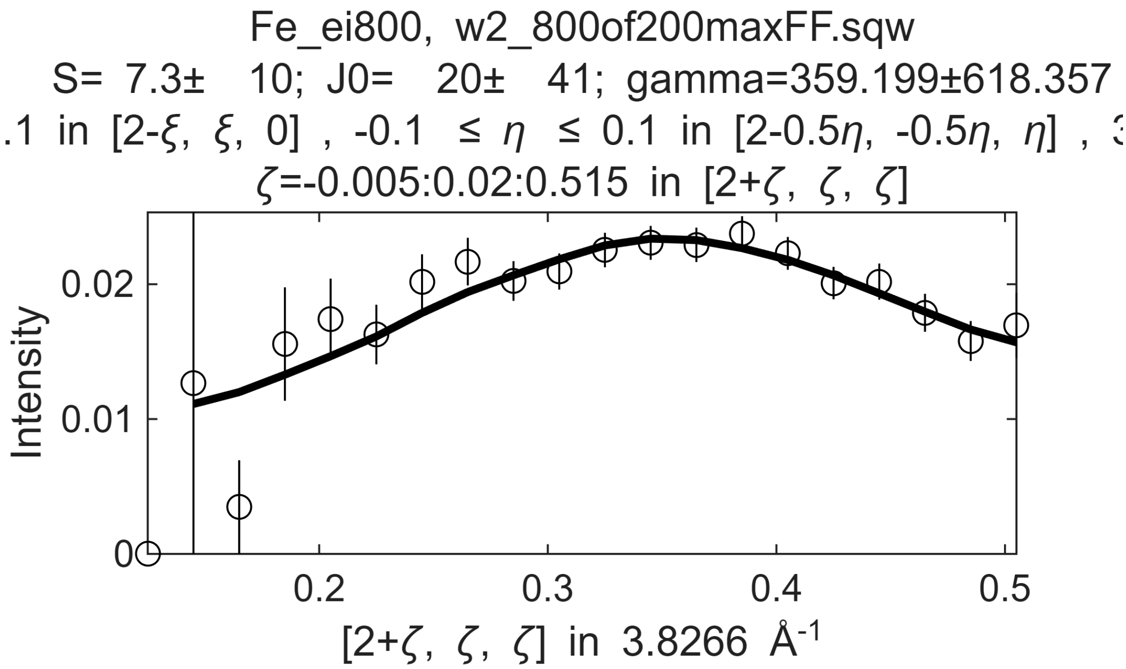


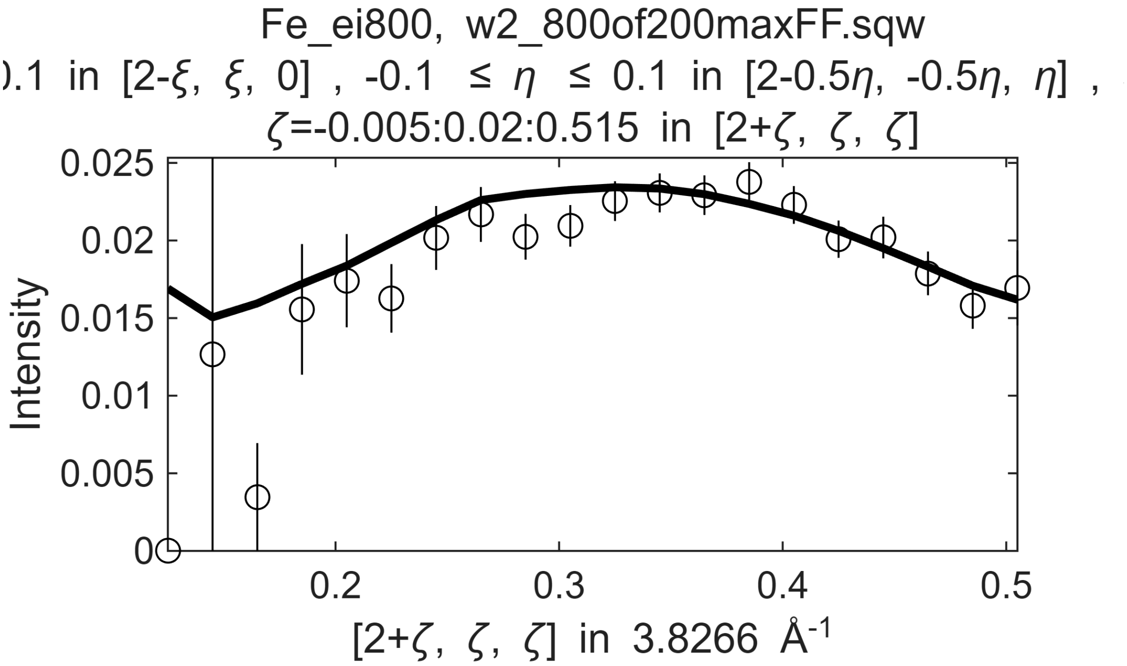

```
******************************
**** En = 345±5 going to 365
******************************
```



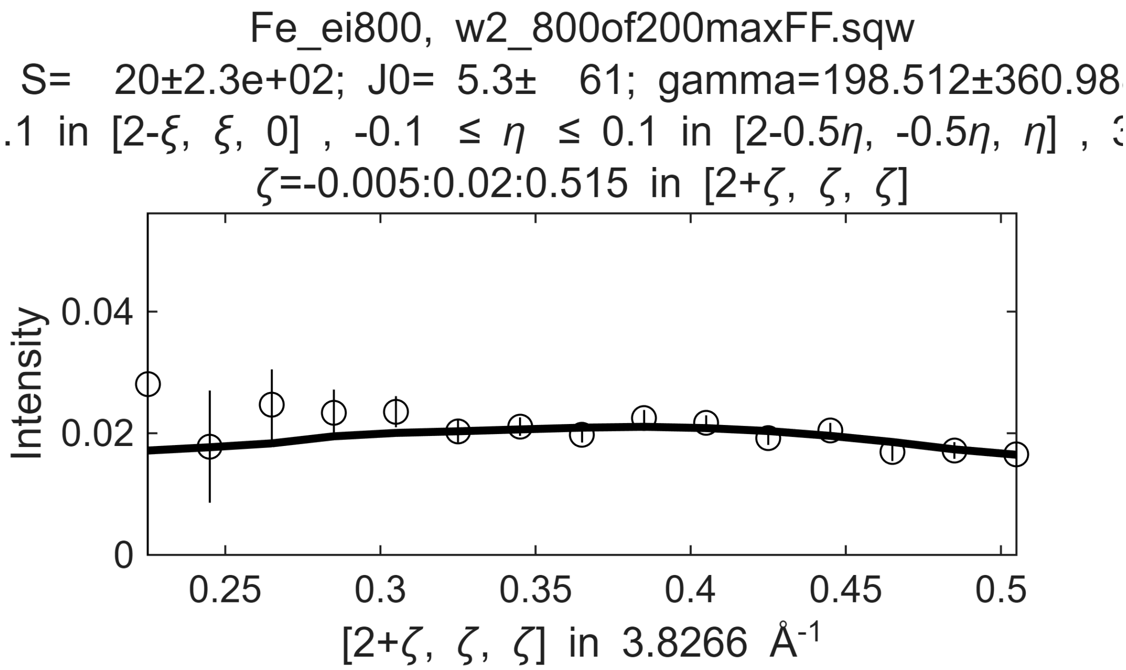


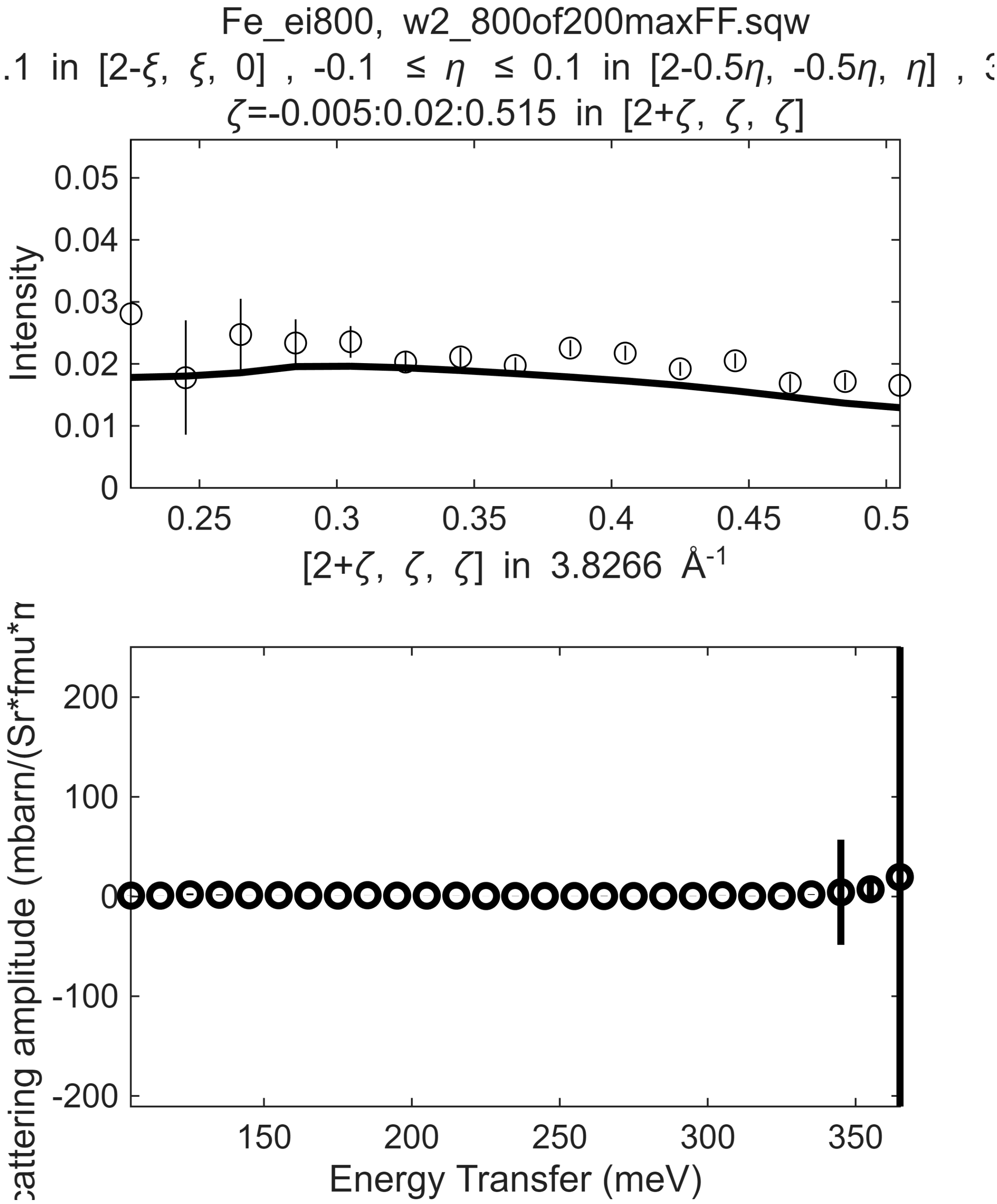

```
******************************
**** En = 355±5 going to 365
******************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 3.8462
```

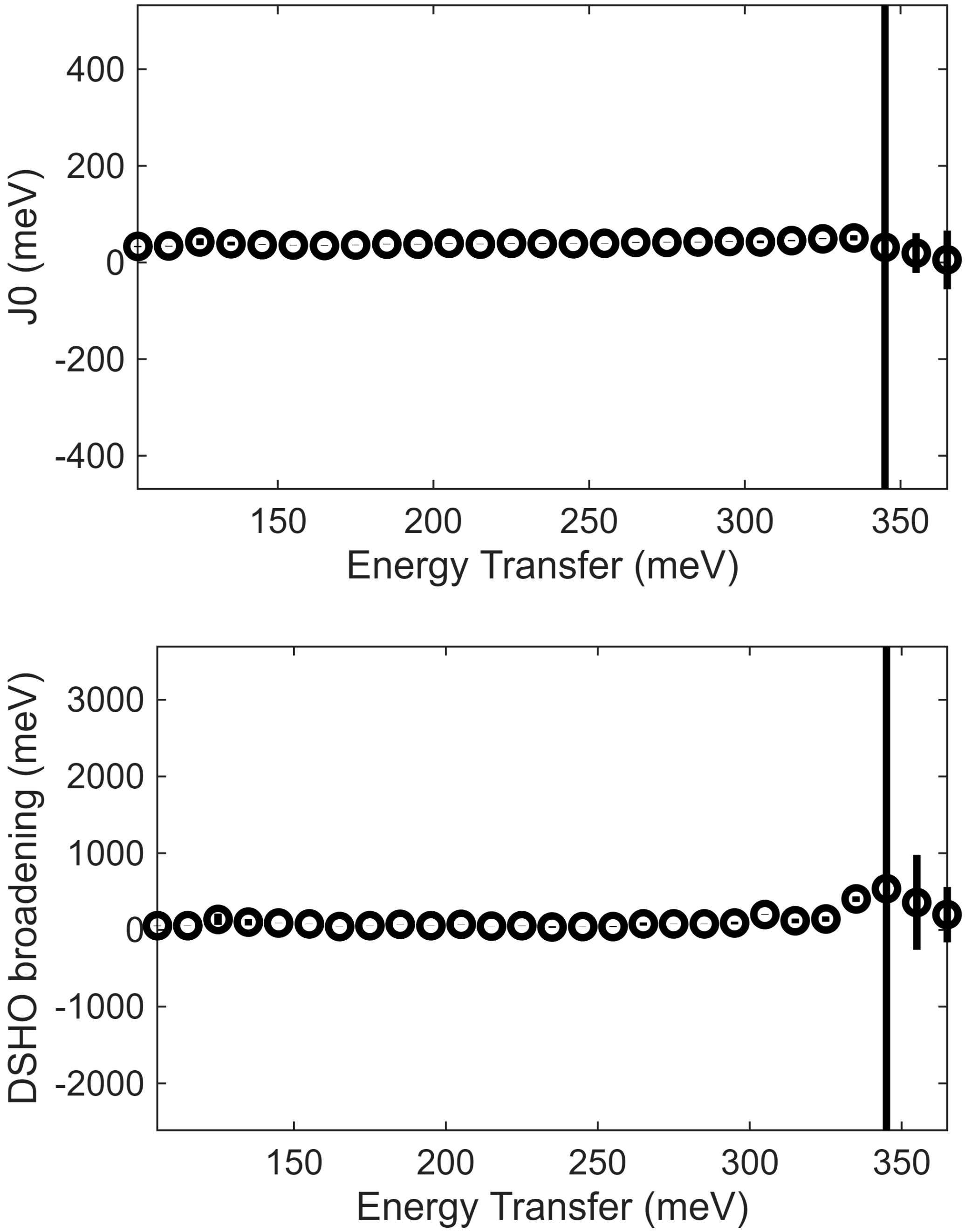


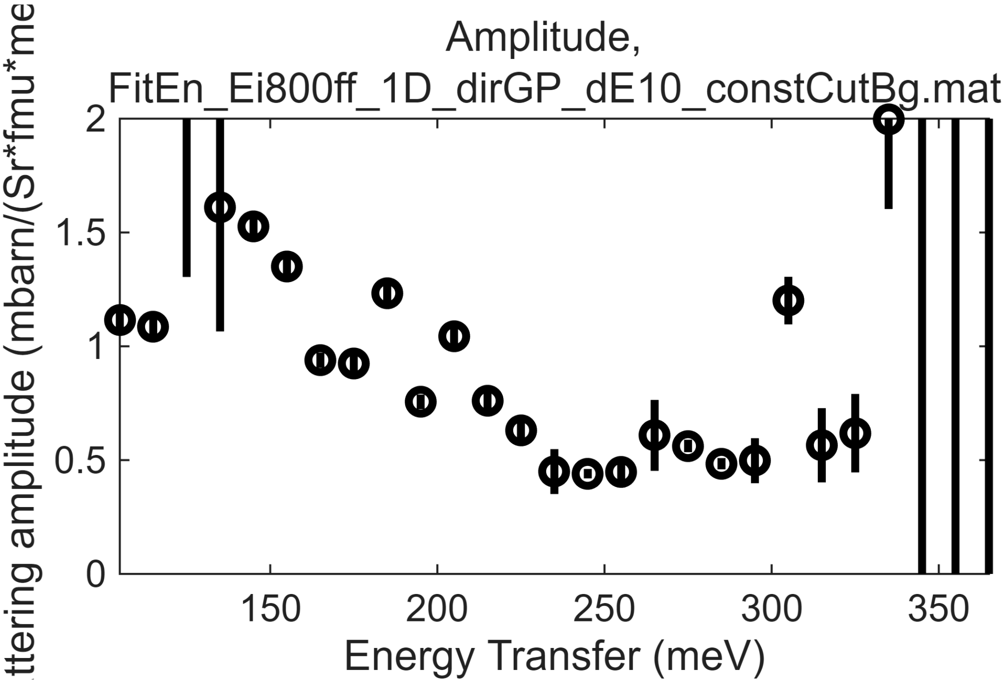


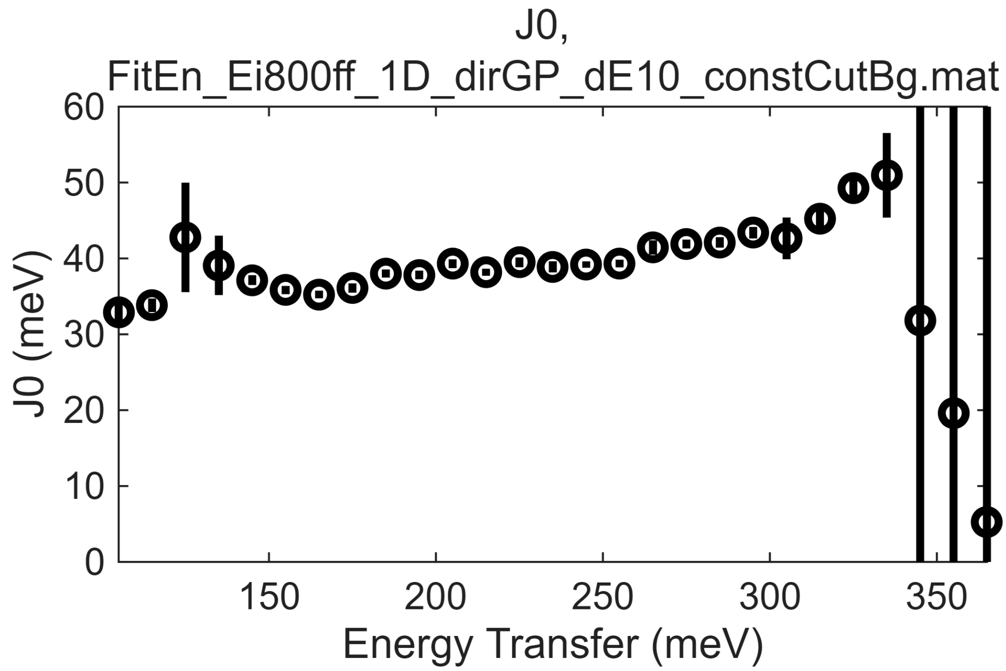

```
******************************
**** En = 365±5 going to 365
******************************
```

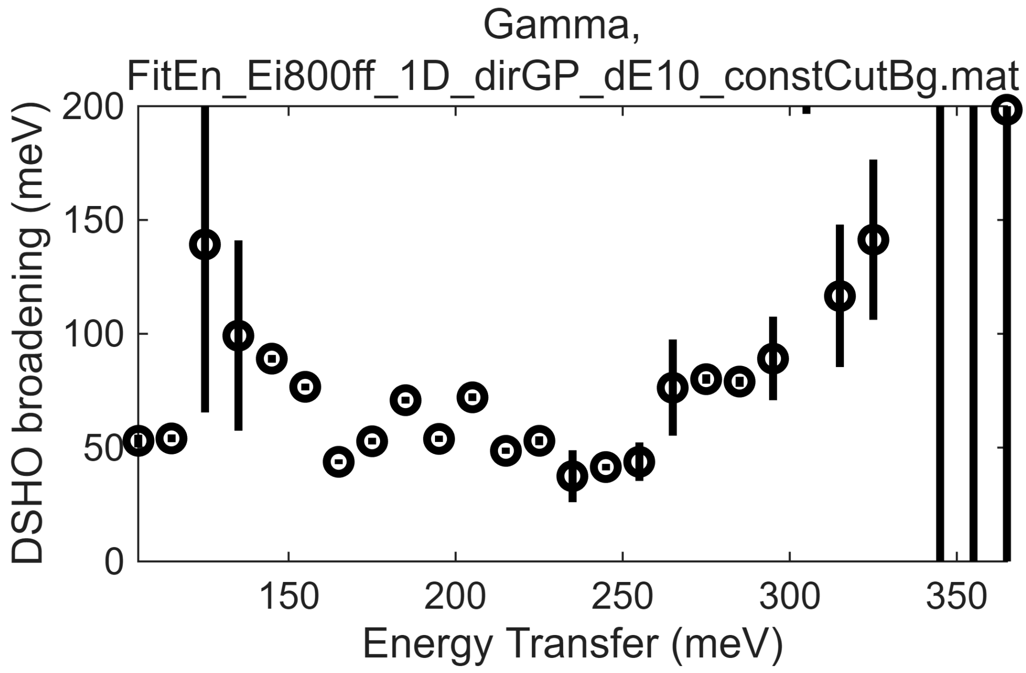


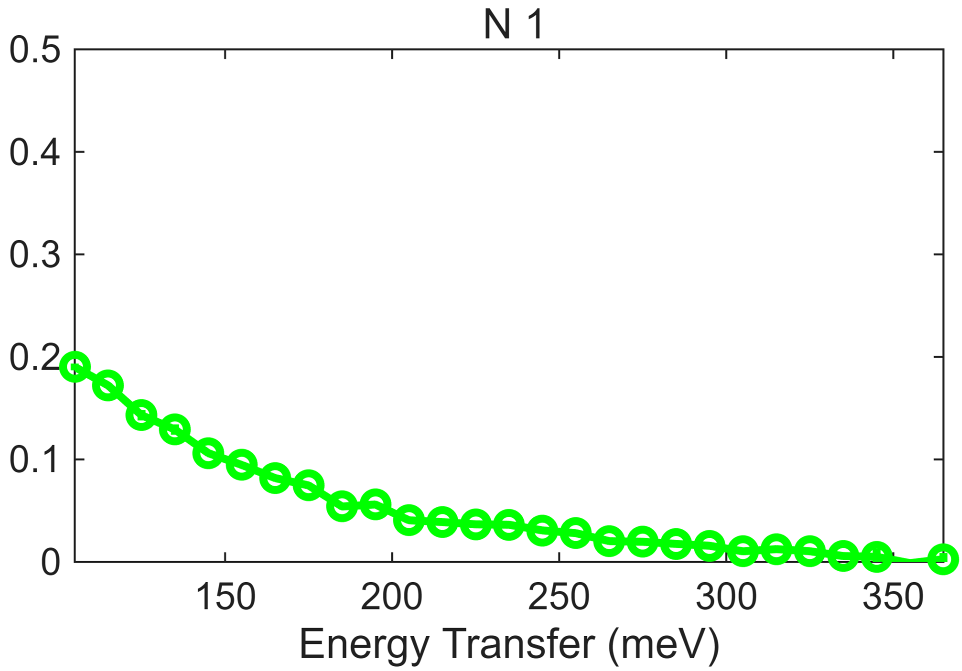

Fe_ei800, w2_800of200maxFF.sqw
.1 in [2-ξ, ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2-0.5η, -0.5η, η] , 3
ζ=-0.005:0.02:0.515 in [2+ζ, ζ, ζ]
Intensity
0.05
0.04
0.03
0.02
0.01
0
0.25
0.3
0.35
0.4
0.45
0.5
[2+ζ, ζ, ζ] in 3.8266 Å-1
cattering amplitude (mbarn/(Sr*fmu*n
200
100
0
-100
-200
150
200
250
300
350
Energy Transfer (meV)

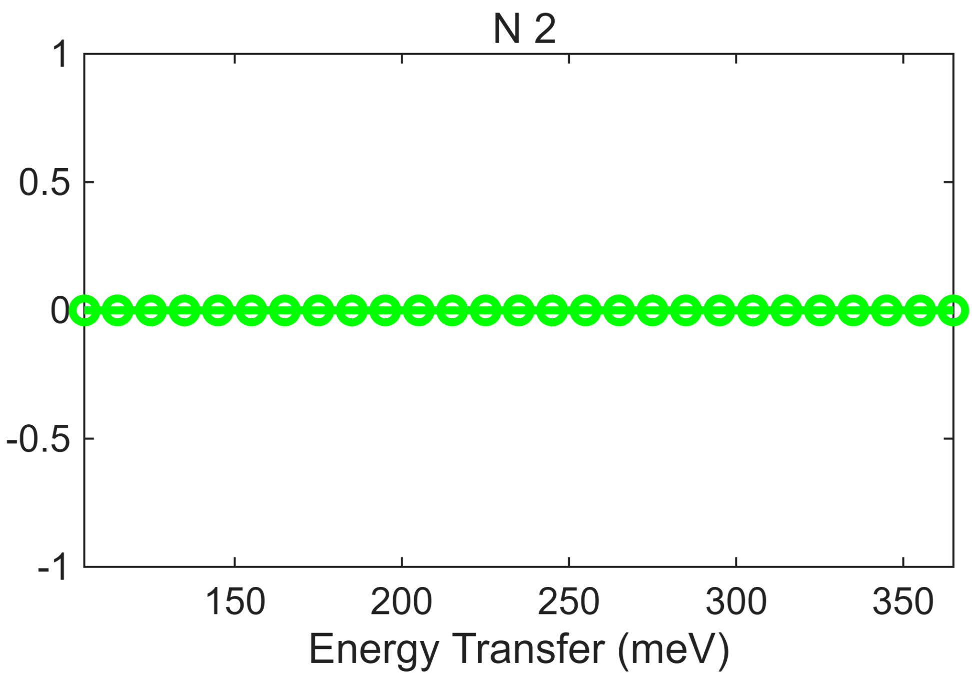

J0 (meV)
400
200
0
-200
-400
150
200
250
300
350
Energy Transfer (meV)
DSHO broadening (meV)
3000
2000
1000
0
-1000
-2000
150
200
250
300
350
Energy Transfer (meV)


Amplitude,
FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat
ttering amplitude (mbarn/(Sr*fmu*me
Energy Transfer (meV)
0
0.5
1
1.5
2
150
200
250
300
350


J0,
FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat
J0 (meV)
Energy Transfer (meV)
0
10
20
30
40
50
60
150
200
250
300
350


Gamma,
FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat
DSHO broadening (meV)
Energy Transfer (meV)


N 1
Energy Transfer (meV)

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******************************
**** CUT: Ei800_off200minFFGP
******************************
******************************
**** CUT: Ei800_off110maxFFGP
******************************
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